Spectroscopy as a Probe of Spacetime Foam

A. G. Kim*
Physics Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA

P. E. Nugent
Applied Mathematics and Computational Research Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA

L. Wang
Department of Physics and Astronomy, 4242 TAMU, College Station, Texas 77843-4242, USA
(Dated: June 16, 2026)

Quantum-gravitational spacetime foam imprints an achromatic, stochastic arrival-time jitter on light from distant sources. We show that this jitter modifies the first-order temporal coherence $g^{(1)}(\tau)$ and hence the observed spectrum, providing a spectroscopic probe complementary to existing imaging-channel (PSF, Strehl) constraints. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a spectral shape-distortion bound on the Ng–Perlman foam exponent α. Applied to the MeV emission lines of GRB 221009A in the dynamical-foam limit ($\tau_{\rm foam} \ll \tau_c$) we obtain $\alpha \gtrsim 0.63$. For the broadband CMB the foam induces an $e^{-\sigma_\phi^2(\nu)}$ deficit ($\sigma_\phi^2 \propto \nu^2$) with a low-frequency pedestal; a fit to the COBE/FIRAS monopole spectrum reaches $\alpha \gtrsim 0.52$ (95% CL) under the physical Planck-scale pedestal kernel, and next-generation spectrometers (PIXIE, SuperPIXIE, Voyage 2050) are forecast to reach $\alpha \gtrsim 0.54$–0.55, the bound saturating logarithmically as the foam deficit is nearly degenerate with a temperature shift. None of these bounds exclude the holographic value $\alpha = 2/3$, which would require a confirmed narrow spectral line at $E_{\rm obs} \gtrsim 720\,{\rm MeV}$ in a $z \sim 0.15$ GRB.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler's original "spacetime foam" picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization due to Ng and collaborators [4–6], the α-foam model, encodes the foam dynamics in a power-law path-length variance

$$\sigma_\ell^2(D) = A_\alpha\, D^{2(1-\alpha)}\, \ell_{\rm P}^{2\alpha}, \tag{1}$$

where D is the line-of-sight comoving distance to the source — the appropriate measure rather than the luminosity distance, since the phase accumulation is a property of spacetime itself rather than the photon path [4] — α parametrizes the foam model, and A_α is a dimensionless coefficient of order unity. A photon traversing distance D passes through approximately $N \sim D/\ell_{\rm P}$ independent Planck-scale spacetime cells, each imparting a tiny random displacement to the photon path; the exponent α controls how the per-cell displacements accumulate: for random-walk foam ($\alpha = 1/2$), contributions add in quadrature ($\sigma_\ell \propto \sqrt{D\ell_{\rm P}}$); holographic foam ($\alpha = 2/3$) grows more slowly, motivated by the heuristic that the information in a spatial region scales with its bounding area rather than its volume — an analogy to the holographic entropy bound, which is rigorous for thermal (black-hole, de Sitter) horizons but is not derived for the random propagation volumes considered here; and the weak-foam limit ($\alpha = 1$) yields a distance-independent $\sigma_\ell \propto \ell_{\rm P}$, corresponding to perfectly correlated cell contributions.

The primary observational probe of stochastic foam to date has been the imaging channel. Light propagating through spacetime foam accumulates a phase dispersion $\Delta\phi \simeq 2\pi\ell^{1-\alpha}\ell_{\rm P}^{\alpha}/\lambda$ across the entrance aperture, degrading

* agkim@lbl.gov

PSF sharpness and erasing interference fringes once $\Delta\phi \gtrsim 2\pi$ [7]. For PKS 1413+135 ($\ell = 1.216$ Gpc) observed with HST at $\lambda = 1.6\,\mu$m, the random-walk model ($\alpha = 1/2$) predicts $\Delta\phi \sim 10 \times 2\pi$, so the detection of Airy rings in this image marginally rules out that model; the holographic model ($\alpha = 2/3$) predicts $\Delta\phi \sim 10^{-9} \times 2\pi$ and is obviously not ruled out [4, 8, 9]. Models with long-range transverse correlations evade even the random-walk bound [10].

The most luminous GRB ever recorded, GRB 221009A at $z = 0.151$ [11], has been examined in the imaging channel by Steinbring [12] and Steinbring & Ng [13]. Using a multiwavelength average of foam-induced angular broadening — analogous to a seeing disk from atmospheric turbulence — Steinbring [12] showed that the holographic ($\alpha = 2/3$) model predicts an angular widening below $\sim 1°$, consistent with the Fermi and Swift localization data for GRBs at $z \leq 5$ up to peak energies of ~ 10 MeV. Steinbring & Ng [13] extended this to the full multiwavelength follow-up of GRB 221009A — from optical and near-infrared through X-rays and the highest-energy gamma-rays ever attributed to a GRB — computing the holographic-foam-widened PSF for real telescopes and finding all data consistent with $\alpha = 2/3$. These imaging-channel analyses show that the holographic model is not excluded by the spatial localization of GRB 221009A.

We develop a framework based on the first-order temporal coherence function $g^{(1)}(\tau)$ of light from astrophysical sources. This function is complementary to the first-order spatial coherence function $g^{(1)}(B)$, which forms the basis of imaging-channel probes. The first-order coherence — the field–field autocorrelation — inherits a foam suppression factor that decreases continuously from unity toward complete fringe erasure as the accumulated phase variance σ_ϕ^2 grows. In our framework the phase variance arises from temporal fluctuations of foam, whereas in the imaging channel it arises from spatial fluctuations; correspondingly, foam imprints itself on the spectrum rather than on the spatial brightness profile.

The remainder of the paper is organized as follows. Section II introduces the spacetime foam model: the arrival-time jitter parameterization, the temporal autocorrelation $C_\alpha(\tau)$, and the distribution of differential jitter. Section III derives the first-order coherence function $g^{(1)}_{\rm obs}(\tau)$ of the observed field and establishes the Gaussian closed form. Section IV translates these results into spectroscopic observables via the Wiener–Khinchin theorem, establishing the split of the observed spectrum into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and a broad pedestal, and connects them to the imaging Strehl ratio. Section V applies the spectral shape-distortion bound to two sources: GRB 221009A (Section V A) and the CMB as observed by COBE/FIRAS (Section V B). Section VI summarizes the main results and outlines future observational directions.

II. SPACETIME FOAM MODEL

A. Arrival-Time and Phase Jitter

This subsection lays out the phenomenological and physical basis for the foam model used throughout the paper. We adopt the Ng–Perlman picture of spacetime foam as a source of stochastic, achromatic path-length fluctuations, and ground that picture in a linearized perturbation of the FLRW background. The resulting kinematic relations fix a single achromatic arrival-time jitter δt, which the observed carrier reads as a phase jitter σ_ϕ and a frequency shift $\delta\omega$ — on which the temporal-correlation and differential-jitter framework developed below is built.

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\rm obs}(t) = E_{\rm src}(t - \delta t(t))\,, \tag{2}$$

where $E_{\rm src}(t)$ is a single spatial mode of a stationary zero-mean complex Gaussian process (chaotic light, appropriate for the thermal and incoherent non-thermal sources considered here, including GRB high-energy emission and the CMB) and $\delta t(t)$ is the foam-induced arrival-time jitter, a stationary stochastic process statistically independent of $E_{\rm src}$. Following the prior foam literature, the source is taken to be a point source along a single line of sight, and $E_{\rm src}(t)$ is the complex field amplitude of that single spatial mode — not a polarization mode. Because the article works entirely at the level of first-order coherence, a spatially extended, incoherent source is recovered as the linear superposition of independent point sources; the superposition is linear in the first-order coherence precisely because the cross terms between mutually incoherent point sources vanish. This is the spatial/angular counterpart of the incoherent narrowband-mode superposition used for the CMB in Section V B 2, and the transverse behavior of the foam across an extended source lies outside the scope of the present treatment. Here t is the observer's time, with the deterministic light-travel delay and cosmological time dilation absorbed into the argument of $E_{\rm src}$, so $\delta t(t)$ is the zero-mean foam fluctuation about the mean arrival time. The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D:

$$\delta t(t) = \frac{1}{c}\,\Delta\ell_{\rm tot}(t). \tag{3}$$

The vacuum c in Eq. (3) is equivalent to propagation through an effective refractive index $n(\tau, r)$, with $\delta t = \frac{1}{c}\int (n-1)\, dr$; the construction of $n(\tau, r)$ in Eqs. (4)–(6) below shows constructively that $n-1$ is geometric and the prefactor is the vacuum c. The key result, established there, is that this path-length shift — and hence δt itself — is achromatic (frequency-independent). To ground this claim, we treat the foam as a stochastic, time-dependent perturbation of a radial Friedmann–Lemaître–Robertson–Walker background. For a radial ray in conformal time τ and comoving radius r,

$$ds^2 = a^2(\tau)\big[-(1+h_{00})\, d\tau^2 + (1+h_{11})\, dr^2\big], \tag{4}$$

with $h_{ab}(\tau, r)$ zero-mean, statistically stationary metric perturbations encoding the Planck-scale foam. The null condition $ds^2 = 0$ gives, to linear order, $dr/d\tau = \sqrt{(1+h_{00})/(1+h_{11})} \simeq 1 + \frac{1}{2}(h_{00} - h_{11})$, so the ray propagates with an effective refractive index

$$n(\tau, r) \;=\; \left(\frac{dr}{d\tau}\right)^{-1} \;\simeq\; 1 - \tfrac{1}{2}\,(h_{00} - h_{11}). \tag{5}$$

As anticipated, $n-1$ is geometric — a property of the metric, not the photon energy — and hence frequency-independent ($n - 1 \not\propto \lambda^2$, unlike a plasma), confirming that the resulting path-length shift is achromatic and distinguishing foam from plasma dispersion, interstellar scintillation, atmospheric turbulence, and (for broadband sources such as the CMB) the standard μ/y spectral distortions, all of which are chromatic. As h_{ab} is zero-mean, $\langle n \rangle = 1$ and the stray $n < 1$ excursions average out: only $\langle (n-1)^2 \rangle$ enters the observables. The arrival-time jitter is then the excess optical path,

$$\delta t \;=\; \frac{1}{c}\int (n-1)\, dr \;=\; -\frac{1}{2c}\int (h_{00} - h_{11})\, dr, \tag{6}$$

identifying $\Delta\ell_{\rm tot} = -\frac{1}{2}\int (h_{00} - h_{11})\, dr$ in Eq. (3). The integral runs over comoving radius, so $\Delta\ell_{\rm tot}$ is a comoving path-length fluctuation with variance $\sigma_\ell^2(D_C) = \langle \Delta\ell_{\rm tot}^2 \rangle$ supplied by Eq. (1). The arrival-time jitter is thus fixed entirely by the metric and is achromatic: it is the same delay for every wavelength.

The phase jitter follows from the arrival-time jitter by the elementary relation between path and phase. The foam is a single achromatic path-length fluctuation $\delta\ell = \int (n-1)\, dr$, which the carrier reads as a phase through the observer-frame wavenumber $2\pi/\lambda_{\rm obs}$. The accumulated foam phase is therefore the line-of-sight integral of the achromatic path-excess density $n-1$ — the per-unit-length integrand of Eq. (6) — at the bare carrier wavenumber,

$$\delta\phi \;=\; \frac{2\pi}{\lambda_{\rm obs}} \int_0^{D_C} \big[n(\tau, r) - 1\big]\, dr \;=\; \omega_0\, \delta t, \tag{7}$$

the phase analogue of the delay integral Eq. (6). Here the comoving radius r parametrizes the line of sight and supplies the true line-of-sight measure dr, while the accumulated path excess $\delta\ell = \int (n-1)\, dr$ enters only through its rate of accumulation $d(\delta\ell)/dr = n - 1$ along the ray. The delay δt and the phase $\delta\phi$ are the group (envelope) delay and the carrier phase of a single propagation, $E_{\rm obs}(t) = e^{-i\omega_0 t}\, e^{\,i\delta\phi(t)}\, A(t - \delta t)$ with A the slow source envelope (derived in Appendix A); the foam is a single achromatic path-length fluctuation, so $\delta\phi = \omega_0\, \delta t$ identically. The time-warp model $E_{\rm obs}(t) = E_{\rm src}(t - \delta t)$ of Eq. (2) carries this same carrier phase, exact for the delay–frequency map of this subsection and in Section III.

Squaring and accumulating the statistically independent per-shell contributions gives the phase variance

$$\sigma_\phi^2 \;=\; \left(\frac{2\pi}{\lambda_{\rm obs}}\right)^2 \sigma_\ell^2(D_C), \tag{8}$$

the phase-pillar analogue of the delay variance $\sigma_\ell^2(D_C)$ of Eq. (1) (with $A_\alpha = 1$), so that $\sigma_\phi = (2\pi/\lambda_{\rm obs})\, \sigma_\ell(D_C)$.

The arrival-time jitter, the phase jitter, and the frequency shift are not three independent effects but the one delay δt read three ways. The delay itself fixes the kinematic map from the metric perturbation h_{ab} to the observable. The carrier reads the same delay as a phase, $\delta\phi = \omega_0\, \delta t$, whose variance σ_ϕ governs the suppression of every observable below; and as an instantaneous frequency, whose excursion is the third: with the observed field $E_{\rm obs}(t) = E_{\rm src}(t - \delta t(t))$ of Eq. (2), a monochromatic carrier acquires the instantaneous frequency

$$\frac{\delta\omega}{\omega_0} \;=\; -\,\frac{d\,\delta t}{dt}, \tag{9}$$

exact for the time-warped carrier of Eq. (2): the phase of $E_{\rm src}(t - \delta t(t))$ advances as $-\omega_0[t - \delta t(t)]$, giving an instantaneous frequency $\omega_0\,(1 - d\,\delta t/dt)$, i.e. Eq. (9) — equivalently, the leading-order perturbation of the geodesic

frequency $\omega = -k_\mu u^\mu$ beyond the background redshift $\omega \propto a^{-1}$. The three projections are one field seen three ways, and the frequency shift is both a consequence of and the evidence for *dynamical* foam. A static screen ($d\,\delta t/dt = 0$) shifts no frequency and leaves the spectrum untouched; only a time-varying delay produces a frequency excursion, with $\langle\delta\omega\rangle = 0$ and a variance set by the temporal structure of the jitter. Here Section II A only announces this observable; the line broadening that its variance imprints on the observed coherence — modifications (i)–(iii) — is derived in Sec. III B and Sec. IV. The quasi-static limit $\delta\omega \to 0$ is the foil: the entire shape-distortion programme rests on the foam being dynamical, which is why the bounds assume the dynamical regime $\tau_{\rm foam} \ll \tau_c$ made precise in Section II B (with the quasi-static caveats deferred to Section VI). What the metric calculation fixes is this kinematic map; the amplitude and temporal structure of the correlations (Secs. II B–II C) are modeled.

We work in the eikonal (geometric-optics) limit of the wave equation in $n(\tau, r)$, in which the foam enters only through the achromatic phase $\delta\phi = \omega_0\,\delta t$ it imprints on the carrier (Eq. (7)). This reduction has a definite range of validity: with $\tau_{\rm foam} \sim \ell_{\rm P}/c$ and foam structure on scales $\ell_{\rm P} \ll \lambda$, the variation is strongly non-adiabatic, and a complete wave treatment would add diffractive and inelastic effects beyond this linear phase screen, not modeled here. The achromaticity established above holds within this reduction; whether it survives a full sub-wavelength treatment ($\ell_{\rm P} \ll \lambda$) is beyond our scope.

B. Temporal Decorrelation of Spacetime Foam

The scalar variance $\sigma_\ell^2(D)$ (Eq. (1)) characterizes the amplitude of foam-induced path-length fluctuations along a single ray; here we model how those fluctuations are correlated across time (temporal decorrelation).

The temporal evolution of spacetime foam — an aspect not specified by Eq. (1) — determines how the arrival-time jitter $\delta t(t)$ along a single line of sight is correlated across the lag τ. Quantum foam models posit that each Planck-scale cell along the line of sight fluctuates on a characteristic timescale $\tau_{\rm foam}$, so that the jitter at times t and $t+\tau$ is correlated for $\tau \ll \tau_{\rm foam}$ and essentially independent for $\tau \gg \tau_{\rm foam}$. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the integrated path-length fluctuation $\Delta\ell_{\rm tot}(t)$, i.e. $C_\alpha(\tau; D) \equiv \langle\Delta\ell_{\rm tot}(t)\,\Delta\ell_{\rm tot}(t+\tau)\rangle/\sigma_\ell^2(D)$, which inherits the temporal structure of the per-cell foam dynamics over the $\sim D/\ell_{\rm P}$ cells crossed by the line of sight. Because C_α is defined on the *integrated* path, its temporal shape inherits a residual dependence on D from the way per-cell foam dynamics combine along the line of sight; the explicit D-argument records this dependence. Equivalently,

$$\big\langle \Delta\ell_{\rm tot}(t)\,\Delta\ell_{\rm tot}(t+\tau)\big\rangle = \sigma_\ell^2(D)\,C_\alpha(\tau; D). \tag{10}$$

Strictly, no physical source is pointlike: the observed light samples the foam over the finite emitting region and along the full propagation path, so the underlying foam correlation is spatio-temporal, $C(\mathbf{r}, \tau)$, rather than purely temporal. We therefore read $C_\alpha(\tau; D)$ as an *effective* temporal correlation obtained by coarse-graining $C(\mathbf{r}, \tau)$ over the source's emitting four-volume and propagation path, with τ parametrizing the residual temporal decorrelation that survives this spatial averaging. The reduction to a purely temporal C_α is exact under the assumption that the spatio-temporal correlation approximately factorizes, $C(\mathbf{r}, \tau) \approx C_{\rm sp}(\mathbf{r})\,C_\alpha(\tau)$, with the spatial factor $C_{\rm sp}(\mathbf{r})$ absorbed into the normalization. We flag this factorization as an assumption rather than a derived result, since the spatial correlation structure over cosmological path lengths is not constrained by the present treatment.

The timescale $\tau_{\rm foam}$ is an additional free parameter of the foam model, not fixed by the α-model variance relation alone. In the standard phenomenological framework [6, 17], each of the $\sim D/\ell_P$ foam cells along the line of sight independently adopts a new configuration on timescale $\tau_P \equiv \ell_P/c \approx 5 \times 10^{-44}$ s; the integrated path-length jitter $\delta\ell_{\rm tot}(t)$ — a sum of $\sim D/\ell_P$ independent contributions — likewise decorrelates on timescale $\tau_{\rm foam} \sim \tau_P$. Because $\tau_{\rm foam} \ll \tau_c$ for any astrophysical source considered here ($\tau_c/\tau_{\rm foam} \gtrsim 10^{22}$ even at 37 MeV), $C_\alpha(\tau)$ has decayed to ≈ 0 over the entire lag range on which the coherence function has support; the foam is thus fully decorrelated on every timescale relevant to the observed spectrum — the most optimistic scenario for detection — and the analysis sits deep within the dynamical regime. Models with long-range *temporal* correlations — the temporal analogue of the long-range *transverse* spatial correlations of Ref. [10] that evade interferometric imaging bounds — would give $C_\alpha(\tau) \approx 1$ over macroscopic timescales and thereby drive $\sigma_\Delta^2(\tau) \to 0$, suppressing the foam signal in both channels analyzed below. The bounds derived here therefore all rest on the dynamical-foam assumption $\tau_{\rm foam} \ll \tau_c$. We carry $C_\alpha(\tau)$ explicitly only because it sets the pedestal width $\sim 1/\tau_{\rm foam}$ (Section IV) and the contrast with the quasi-static limit; for the bounds themselves $C_\alpha \to 0$, an approximation reinforced by transverse decorrelation across the beam (any two rays with $D\,\delta\theta > \ell_P$ are independent), so the fully-decorrelated single-ray treatment is conservative.

C. Differential Jitter and Its Distribution

With the temporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the differential jitter between photons arriving at times t and $t+\tau$:

$$\Delta(\tau) \equiv \delta t(t+\tau) - \delta t(t). \tag{11}$$

Expanding the square and applying Eq. (10):

$$\begin{aligned}\sigma_\Delta^2(\tau) &= \frac{1}{c^2}\big\langle[\Delta\ell_{\rm tot}(t+\tau) - \Delta\ell_{\rm tot}(t)]^2\big\rangle \\ &= \frac{1}{c^2}\left[\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2\, C_\alpha(\tau)\right] \\ &= \frac{2\sigma_\ell^2(D)}{c^2}\left[1 - C_\alpha(\tau; D)\right].\end{aligned} \tag{12}$$

By the central limit theorem (CLT), $\Delta(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta(\tau) \sim \mathcal{N}\big(0,\, \sigma_\Delta^2(\tau)\big)\,, \tag{13}$$

where σ_Δ^2 is taken from Eq. (12); for any source at $D \gtrsim 1$ kpc, the cell count $N_{\rm cells} \sim D/\ell_{\rm P} \gtrsim 10^{54}$ amply satisfies the large-N requirement and higher-cumulant corrections are negligible. For $\alpha = 1/2$ (random walk), per-cell path-length variances are equal ($\ell_{\rm P}^2$ each) and the CLT applies in its standard form. For $\alpha \neq 1/2$, the per-cell contributions scale differently with D and the equality of per-cell variances is a model assumption rather than a direct CLT consequence; Gaussianity is adopted as the minimal assumption consistent with the α-model.

The differential delay $\Delta(\tau)$ and its variance $\sigma_\Delta^2(\tau)$ are the kinematic content of this subsection. The carrier-phase suppression they imprint on the observed coherence — the foam factor $\mathcal{F}_{\rm foam}(\tau)$ — arises only when the carrier enters $g^{(1)}$, and is derived in Section III.

III. FIRST-ORDER COHERENCE

A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g^{(1)}_{\rm obs}(\tau) = \frac{\langle E^*_{\rm obs}(t)\, E_{\rm obs}(t+\tau)\rangle}{\langle|E_{\rm obs}(t)|^2\rangle}. \tag{14}$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t+\tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of $E_{\rm src}$ gives

$$\langle E^*_{\rm src}(u)\, E_{\rm src}(v)\rangle = \langle|E_{\rm src}|^2\rangle\, g^{(1)}_{\rm src}(\tau - \Delta(\tau)). \tag{15}$$

Since δt is independent of $E_{\rm src}$, and since $\langle|E_{\rm obs}(t)|^2\rangle = \langle|E_{\rm src}(t-\delta t)|^2\rangle = \langle|E_{\rm src}|^2\rangle$ by stationarity (so the normalization in Eq. (14) cancels), averaging over the jitter yields

$$g^{(1)}_{\rm obs}(\tau) = \big\langle g^{(1)}_{\rm src}(\tau - \Delta(\tau))\big\rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau))\, g^{(1)}_{\rm src}(\tau - \Delta)\, d\Delta. \tag{16}$$

For a quasi-monochromatic source $g^{(1)}_{\rm src}(\tau) = e^{-i\omega_0\tau}\,\gamma(\tau)$ with γ the slowly varying envelope coherence, Eq. (16) reads

$$g^{(1)}_{\rm obs}(\tau) = e^{-i\omega_0\tau}\,\big\langle e^{\,i\omega_0\Delta(\tau)}\,\gamma(\tau - \Delta(\tau))\big\rangle_{\delta t}, \tag{17}$$

the envelope responding to the achromatic differential delay $\Delta(\tau)$ (variance σ_Δ^2) and the carrier acquiring the phase $\omega_0\Delta(\tau)$. The carrier suppression is the characteristic function of that phase,

$$\mathcal{F}_{\rm foam}(\tau) \equiv \big\langle e^{\,i\omega_0\Delta(\tau)}\big\rangle_{\delta t} = e^{-\omega_0^2\sigma_\Delta^2(\tau)/2} = e^{-\sigma_\phi^2\,[1 - C_\alpha(\tau; D)]}, \tag{18}$$

with $\mathcal{F}_{\rm foam}(0) = 1$, the differential phase variance $D_\phi(\tau) \equiv \omega_0^2\sigma_\Delta^2(\tau) = 2\sigma_\phi^2[1 - C_\alpha(\tau; D)]$, and $\sigma_\phi^2 = (2\pi/\lambda_{\rm obs})^2\sigma_\ell^2(D_C)$ (Eq. (8)); at $\tau \to \infty$ ($C_\alpha \to 0$), $\mathcal{F}_{\rm foam} \to e^{-\sigma_\phi^2}$. Thus the observed first-order coherence is the intrinsic coherence convolved with the Gaussian differential-jitter kernel.

B. Gaussian Source Profile

The GRB 221009A emission line is well described by a Gaussian profile in each epoch [18], and it serves as the representative case admitting the simplest closed form. A Gaussian line of 1σ width σ_ν centered on ν_0 corresponds via the Wiener–Khinchin theorem to the coherence function

$$g^{(1)}_{\rm src}(\tau) \;=\; e^{-i\omega_0\tau}\, e^{-\tau^2/2\tau_G^2}, \tag{19}$$

where $\omega_0 \equiv 2\pi\nu_0$ and $\tau_G \equiv 1/(2\pi\sigma_\nu)$ is the Gaussian coherence decay time. The carrier factor $e^{-i\omega_0\tau}$ encodes the line's position in the spectrum and must be retained: dropping it would substitute a baseband Gaussian centered at $\nu = 0$ for the physical line and would lose the dominant foam effect (line-peak suppression). Substituting the Gaussian profile (19) into the jitter-convolution integral (17) gives, with $\sigma_\Delta^2 \equiv \sigma_\Delta^2(\tau)$,

$$g^{(1)}_{\rm obs}(\tau) = e^{-i\omega_0\tau}\int_{-\infty}^{\infty} \frac{d\Delta}{\sqrt{2\pi\sigma_\Delta^2}}\, \exp\!\left[-\frac{\Delta^2}{2\sigma_\Delta^2} + i\omega_0\Delta - \frac{(\tau-\Delta)^2}{2\tau_G^2}\right]. \tag{20}$$

The exponent is quadratic in Δ; writing it as $-A\,\Delta^2 + B\,\Delta - \tau^2/2\tau_G^2$ with $A = (\tau_G^2+\sigma_\Delta^2)/(2\sigma_\Delta^2\tau_G^2)$ and $B = i\omega_0 + \tau/\tau_G^2$, the Gaussian integral $\int e^{-A\Delta^2+B\Delta}\,d\Delta = \sqrt{\pi/A}\,e^{B^2/4A}$ produces the amplitude prefactor $\tau_G/\sqrt{\tau_G^2+\sigma_\Delta^2}$ together with the exponent $B^2/4A - \tau^2/2\tau_G^2$, whose imaginary, τ^2, and ω_0^2 parts separate into a carrier shift, an envelope broadening, and an amplitude suppression to give the exact closed form (see also Appendix A)

$$\begin{aligned} g^{(1)}_{\rm obs}(\tau) = {}& \frac{\tau_G}{\sqrt{\tau_G^2+\sigma_\Delta^2(\tau)}}\, \exp\!\left[-i\omega_0\tau + \frac{i\,\omega_0\,\sigma_\Delta^2(\tau)\,\tau}{\tau_G^2+\sigma_\Delta^2(\tau)}\right] \\ & \times \exp\!\left[-\frac{\tau^2}{2\big[\tau_G^2+\sigma_\Delta^2(\tau)\big]}\right] \exp\!\left[-\frac{D_\phi(\tau)}{2} + \frac{\omega_0^2\sigma_\Delta^4(\tau)}{2\big[\tau_G^2+\sigma_\Delta^2(\tau)\big]}\right]. \end{aligned} \tag{21}$$

The observed coherence is again a Gaussian-modulated oscillation, but with three foam-induced modifications: (i) a slight shift of the carrier, $\nu_0^{\rm eff} = \nu_0\big[1-\sigma_\Delta^2(\tau)/(\tau_G^2+\sigma_\Delta^2(\tau))\big]$; (ii) a wider Gaussian decay of $g^{(1)}_{\rm obs}(\tau)$ in lag (decay scale $\sqrt{\tau_G^2+\sigma_\Delta^2(\tau)}$ vs. τ_G); and (iii) a global amplitude suppression set by the differential phase variance $D_\phi(\tau) = \omega_0^2\sigma_\Delta^2(\tau)$ (Eq. (18)), at the carrier ω_0. The marginal phase variance $D_\phi = \omega_0^2\sigma_\Delta^2$ is exact; the carrier shift (i) and the $\omega_0^2\sigma_\Delta^4$ amplitude term are the carrier–envelope coupling, both of order σ_Δ^2/τ_G^2 and vanishing in the slowly-varying-envelope limit $\sigma_\Delta \ll \tau_G$, where $\mathcal{F}_{\rm foam} = e^{-D_\phi/2}$ factors out cleanly, $g^{(1)}_{\rm obs} \to \mathcal{F}_{\rm foam}(\tau)\, g^{(1)}_{\rm src}(\tau)$ — the pure multiplicative suppression on which the spectroscopic bounds rest. Normalization $g^{(1)}_{\rm obs}(0) = 1$ holds because $\sigma_\Delta(0) = 0$ (hence $D_\phi(0) = 0$). The narrowband limit $\tau_G \to \infty$ of Eq. (21) reduces to a pure phase-modulated oscillation, which is the single-mode building block from which the broadband CMB spectrum of Section V B 2 is constructed by incoherent superposition.

IV. IMPLICATIONS FOR SPECTROSCOPY

A. The Wiener–Khinchin Bridge

The foam-modified observed coherence function — given in general by the jitter-convolution integral Eq. (16) with the closed form Eq. (21) — carries direct spectroscopic content: for a stationary field the power spectral density $S(\nu)$ and $g^{(1)}(\tau)$ are Fourier-transform pairs, implying

$$S_{\rm obs}(\nu) = \mathcal{F}\Big\{g^{(1)}_{\rm obs}(\tau)\Big\}(\nu). \tag{22}$$

The theorem applies because astronomical emission lines have finite linewidth (so the Fourier transform is well defined) and because the foam-induced jitter is a stationary process independent of the source (foam acts on light after emission, while $\delta t(t)$ statistics are time-translation invariant on observational timescales). The observed field is then stationary when the integration time exceeds the foam coherence time $\tau_{\rm foam}$ (the lag scale beyond which $C_\alpha(\tau; D) \to 0$), so that the time-averaged spectrum coincides with its ensemble average: $T_{\rm int} \gg \tau_{\rm foam}$.

In the dynamical limit $C_\alpha(\tau; D) \to 0$ for $|\tau| \gtrsim \tau_{\rm foam}$, $\mathcal{F}_{\rm foam}(\tau)$ acquires a long-lag plateau $e^{-\sigma_\phi^2}$ outside a narrow window $|\tau| \lesssim \tau_{\rm foam}$. Fourier-transforming via Eq. (22), the plateau produces a coherent core $e^{-\sigma_\phi^2} S_{\rm src}(\nu)$, while the

short-lag window redistributes the complementary fraction $1 - e^{-\sigma_\phi^2}$ of the source flux into a broad pedestal. This core-plus-pedestal split holds for any source, with σ_ϕ^2 evaluated frequency by frequency ($\sigma_\phi^2 \to \sigma_\phi^2(\nu)$) for a broadband source; its broadband form for the CMB, including the pedestal shape, is worked out for a specific model of the temporal autocorrelation C_α in Section V B 2.

For the Gaussian source profile of Section III B, Fourier-transforming Eq. (21) with σ_Δ^2 held at its asymptotic value $2\sigma_\ell^2/c^2$ gives the explicit Gaussian core

$$S_{\rm obs}^{\rm core}(\nu) \;=\; \frac{\tau_G}{\sqrt{\tau_G^2+\sigma_\Delta^2}}\,\exp\!\left[-\frac{\sigma_\phi^2\,\tau_G^2}{\tau_G^2+\sigma_\Delta^2}\right]\cdot\frac{1}{\sqrt{2\pi}\,\sigma_{\rm obs}}\exp\!\left[-\frac{(\nu-\nu_0^{\rm eff})^2}{2\sigma_{\rm obs}^2}\right], \tag{23}$$

with $\nu_0^{\rm eff} = \nu_0\tau_G^2/(\tau_G^2+\sigma_\Delta^2)$ and $\sigma_{\rm obs} = 1/[2\pi\sqrt{\tau_G^2+\sigma_\Delta^2}] < \sigma_\nu$; the result is exact for the single-carrier Gaussian source of Eq. (21). The Gaussian's carrier shift, line narrowing, and peak suppression are the frequency-domain images of items (i)–(iii) of Section III B.

For completeness, the coherent core (23) carries only the fraction $e^{-\sigma_\phi^2}$ of the line flux; the complementary $1 - e^{-\sigma_\phi^2}$ reappears as a pedestal. In the relevant regime $\sigma_\Delta \ll \tau_G$ the closed form (21) factorizes, $g_{\rm obs}^{(1)}(\tau) = e^{-i\omega_0\tau}e^{-\tau^2/2\tau_G^2}\,\mathcal{F}_{\rm foam}(\tau)$ with $\mathcal{F}_{\rm foam}(\tau) = e^{-\sigma_\phi^2[1-C_\alpha(\tau)]}$, and the plateau/residual split $\mathcal{F}_{\rm foam}(\tau) = e^{-\sigma_\phi^2} + [e^{-\sigma_\phi^2[1-C_\alpha(\tau)]} - e^{-\sigma_\phi^2}]$ used in the broadband case [Eq. (38)] sends the plateau to the core (23) and the short-lag residual to the pedestal

$$\begin{aligned} S_{\rm obs}^{\rm ped}(\nu) &= \int_{-\infty}^{\infty} d\tau\, e^{\,i2\pi(\nu-\nu_0)\tau}\, e^{-\tau^2/2\tau_G^2}\left[e^{-\sigma_\phi^2[1-C_\alpha(\tau)]} - e^{-\sigma_\phi^2}\right] \\ &\xrightarrow{\tau_{\rm foam}\ll\tau_G} \int_{-\infty}^{\infty} d\tau\, e^{\,i2\pi(\nu-\nu_0)\tau}\left[e^{-\sigma_\phi^2[1-C_\alpha(\tau)]} - e^{-\sigma_\phi^2}\right] \equiv \hat{R}(\nu-\nu_0), \end{aligned} \tag{24}$$

so that the full observed line is $S_{\rm obs}(\nu) = S_{\rm obs}^{\rm core}(\nu) + S_{\rm obs}^{\rm ped}(\nu)$. The residual is sourced by the short-lag window $|\tau| \lesssim \tau_{\rm foam}$, over which the envelope $e^{-\tau^2/2\tau_G^2} \approx 1$; the resulting first-order kernel $\hat{R}(\nu-\nu_0)$ is the line-centered image of the broadband residual of Eq. (38), spread over $\sim 1/\tau_{\rm foam}$. Its weight $\int S_{\rm obs}^{\rm ped}\,d\nu = 1 - e^{-\sigma_\phi^2}$ exactly complements the core's $e^{-\sigma_\phi^2}$, conserving the line photon number. When $\tau_{\rm foam} \ll \tau_G, \sigma_\Delta$ the pedestal is correspondingly broad and shallow, and the spectrum near the line peak is dominated by the coherent core.

B. Connection to the Imaging Strehl Ratio

The first-order coherence function $g_{\rm obs}^{(1)}(\tau)$ derived above describes *temporal* coherence along a single line of sight, with the foam suppression governed by the temporal decorrelation $C_\alpha(\tau; D)$ of Eq. (18). The Strehl ratio of an imaging telescope is the *spatial* counterpart: phase variations across the aperture plane are governed by an aperture-plane coherence function $C_\alpha^{\rm sp}(B; D)$, constructed by direct analogy to the temporal $C_\alpha(\tau; D)$ of Section II B under the same foam-cell statistics.

The structural equivalence is not merely formal: in both cases the foam suppression arises as the characteristic function of a zero-mean Gaussian differential random variable. Temporally, the variable is the differential jitter $\Delta(\tau) = \delta t(t+\tau) - \delta t(t)$ with variance $\sigma_\Delta^2(\tau) = (2\sigma_\ell^2/c^2)[1 - C_\alpha(\tau; D)]$ (Eq. (12)). Spatially, the variable is the differential phase $\Delta\phi(B) = \phi(\mathbf{x}+\mathbf{B}) - \phi(\mathbf{x})$ between two points on the aperture plane separated by baseline B, with variance

$$\sigma_{\Delta\phi}^2(B) \;=\; 2\sigma_\phi^2\big[1 - C_\alpha^{\rm sp}(B; D)\big]. \tag{25}$$

The focal-plane field of a telescope is the Fraunhofer Fourier transform of the aperture-plane field, giving the diffraction-limited PSF when the aperture phase is uniform. Foam imprints a Gaussian random phase $\phi(\mathbf{x})$ on the aperture with variance σ_ϕ^2 and two-point correlation $C_\alpha^{\rm sp}(B; D)$, blurring this PSF. In the resolved limit $C_\alpha^{\rm sp}(D_{\rm tel}; D) \to 0$ — where aperture phases at distinct points decorrelate — the on-axis peak is attenuated by the Strehl ratio

$$S \;\simeq\; e^{-\sigma_\phi^2} \tag{26}$$

(extended-Maréchal [4]), the surrounding Airy fringes are washed out by the same factor, and the complementary $1 - e^{-\sigma_\phi^2}$ of the source flux scatters into an incoherent halo.

Imaging and spectroscopic channels each probe σ_ϕ^2 through a Fourier transform — aperture position $\leftrightarrow$ focal-plane angle in imaging (Fraunhofer), lag $\leftrightarrow$ frequency in spectroscopy (Wiener–Khinchin) — with the rapid chromatic

tightening $\sigma_\phi \propto 1/\lambda$ exploited in both Ng & Perlman's [4] Table 1 and the analyses of Section V A. The shared σ_ϕ^2 that suppresses the coherent diffraction-limited PSF in imaging also suppresses the coherent line core in spectroscopy, placing the published imaging constraints [4, 7–9] on a common footing with the present framework. A quantitative comparison across channels is, however, model-dependent: it requires relating the spatial ($C_\alpha^{\rm sp}$) and temporal (C_α) foam correlation functions under a common foam-cell model, so the two constraints coincide only to the precision of those shared model details. The duality is not perfect at the observational interface: in the imaging channel the reference truth—the telescope aperture—is known a priori, whereas in the spectroscopic channel the corresponding truth—the intrinsic source spectrum—is itself uncertain, so the foam contribution is harder to isolate.

V. SOURCES

Time-varying spacetime foam modifies the source first-order coherence, and hence its spectrum, in a wavelength-dependent way through the scaling $\sigma_\phi \propto 1/\lambda$. Two observable consequences follow. First, individual spectral features are distorted according to the Gaussian-source result Eq. (23), with a carrier shift, line narrowing, and peak suppression that scale chromatically as $\sigma_\phi \propto 1/\lambda$. Second, the broadband spectral energy distribution is distorted: the cosmic microwave background, expected to be a near-perfect blackbody, would acquire foam-induced deviations whose chromatic signature is distinct from standard chemical-potential and Compton-y distortions. In this section we consider these two possibilities.

A. GRB 221009A

GRB 221009A affords two independent foam probes that draw on the same physics but differ sharply in strength and in what they assume. The stronger of the two is an *absolute* one: time-varying foam suppresses the coherent core of any source spectrum by $e^{-\sigma_\phi^2(\nu)}$ (Eq. (18)), and because the complementary flux is redistributed into a pedestal of enormous bandwidth, essentially none of it returns to the observing band. The in-band flux is therefore attenuated outright, so the mere survival of the TeV photons detected by LHAASO already bounds σ_ϕ^2 at TeV energies and hence α. The weaker, secondary probe is a *relative* one: the energy-dependent broadening of the MeV emission line, which is insensitive to the absolute suppression and constrains α only in the deep strong-foam regime. We develop the absolute bound first and demote the line-width analysis to a cross-check, because the former is both more stringent and rests on weaker assumptions.

GRB 221009A ("the BOAT," Brightest Of All Time) is the most energetic gamma-ray burst in the instrumental record, with isotropic-equivalent energy $E_{\rm iso} \sim 10^{55}\,$erg at redshift $z = 0.151$ [11], a once-in-$\sim 10^4$-year event [19]. Adopting the Planck 2018 flat ΛCDM cosmology used throughout [20] ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.36\,{\rm km\,s^{-1}\,Mpc^{-1}}$), the comoving distance is $D_C(z = 0.151) \approx 0.65\,{\rm Gpc} \approx 2.0 \times 10^{25}\,{\rm m}$ and $\ln(D_C/\ell_P) \approx 138.4$. Both probes below operate in the dynamical-foam regime $\tau_{\rm foam} \ll \tau_c$ adopted throughout Section II; the opposite quasi-static limit is addressed explicitly as Assumption (i) below.

1. *Mechanism and assumptions*

The foam-modified spectrum of a broadband source splits into a coherent core and an incoherent pedestal,

$$S_{\rm obs}(\nu) = e^{-\sigma_\phi^2(\nu)}\, B_\nu + \int B_{\nu'}\, \hat{R}(\nu - \nu'; \nu')\, d\nu', \tag{27}$$

derived in full for the blackbody as Eq. (38) but valid for any source: the first term is the surviving carrier, suppressed per mode by $e^{-\sigma_\phi^2(\nu)}$, while the second is the flux removed from the core, $1 - e^{-\sigma_\phi^2}$, redistributed by the pedestal kernel $\hat{R}$ of width $\sim 1/\tau_{\rm foam}$. The decomposition conserves photon number *globally*, integrated over the full pedestal, but not within any finite observing band; whether the redistributed flux returns to the band is the crux of what follows, and is governed by two assumptions.

Assumption (i): dynamical foam. The pedestal exists only if the foam decorrelates on accessible timescales, $C_\alpha(\tau) \to 0$; in the quasi-static limit $C_\alpha \to 1$ the foam factor $\mathcal{F}_{\rm foam} \to 1$ and there is no suppression to constrain (Section II B). This escape is closed at TeV energies by the paper's own estimate: with $\tau_{\rm foam} \sim \tau_P \equiv \ell_P/c \approx 5 \times 10^{-44}\,$s, the foam is already dynamical with margin $\tau_c/\tau_{\rm foam} \gtrsim 10^{22}$ *even at* $37\,MeV$ (Section II B), and since the coherence time $\tau_c \propto 1/\nu$ shrinks with energy, the margin only grows at TeV. Whatever sets the foam dynamical at MeV sets it more dynamical at TeV, so the quasi-static loophole cannot be invoked selectively to spare the TeV bound.

Assumption (ii): the source bandwidth is far narrower than the pedestal. The pedestal width is $\sim 1/\tau_{\rm foam} \sim c/\ell_P \approx 10^{43}$ Hz, whereas the entire electromagnetic output of a GRB — radio to TeV — spans $\lesssim 10^{27}$ Hz. The ratio of source bandwidth to pedestal bandwidth is $\sim 10^{-16}$, and this inequality holds for *every* astrophysical source, the CMB included. Relative to the kernel the whole source is effectively a single line, so the flux scattered out of the core is spread over ~ 16 decades of frequency and the fraction returning to any astrophysical band is $\sim 10^{-16}$ — negligible.

2. Absolute in-band suppression

With the in-band pedestal negligible by Assumption (ii), the second term of Eq. (27) drops inside the band and the observed in-band flux reduces to

$$S_{\rm obs}(\nu)\Big|_{\rm in\text{-}band} \;\approx\; e^{-\sigma_\phi^2(\nu)}\, S_{\rm src}(\nu), \tag{28}$$

an *absolute* attenuation of the source flux that $\to 0$ as $\sigma_\phi^2(\nu) \gg 1$. We attach the $\to 0$ to the observed in-band flux of Eq. (28), not to the bare factor $e^{-\sigma_\phi^2}$ in Eq. (27): photon number is conserved globally, and it is only after the pedestal has been shown to escape the band that the surviving in-band flux genuinely vanishes. This is the key distinction from single-feature spectroscopy. The statement that "uniform dimming is undetectable" — the smooth $e^{-\sigma_\phi^2(\nu)}$ attenuates line and continuum equally at a given frequency, preserving the equivalent width — holds only for *relative* measurements within one spectrum, where no external flux reference is available. It does *not* protect an *absolute* flux: a measurement that compares the detected flux to an independently predicted intrinsic flux is sensitive to the suppression directly.

3. Primary bound: survival of the LHAASO TeV photons

LHAASO detected GRB 221009A in TeV gamma rays [21]. By Eq. (28) the observed in-band TeV photon count is $N \approx e^{-\sigma_\phi^2(E_{\rm TeV})}\, N_0$, with N_0 the count expected from the modeled intrinsic fluence after standard extragalactic-background-light attenuation. A detection of N photons against the prediction N_0 therefore requires

$$e^{-\sigma_\phi^2(E_{\rm TeV})} \;\gtrsim\; \frac{N}{N_0}, \qquad \Longrightarrow \qquad \sigma_\phi^2(E_{\rm TeV}) \;\lesssim\; \ln\frac{N_0}{N} \;\equiv\; \sigma^2_{\phi,\max} \;=\; \mathcal{O}(\text{few}), \tag{29}$$

since even a complete intrinsic model leaves N/N_0 within a small factor of unity.[VERIFY] Writing $\sigma_\phi^2(E_{\rm TeV}) = (2\pi/\lambda_{\rm obs})^2 D_C^{2(1-\alpha)} \ell_P^{2\alpha} = (2\pi D_C/\lambda_{\rm obs})^2 (\ell_P/D_C)^{2\alpha}$ from Eqs. (1) and (8) (with $A_\alpha = 1$ and $\lambda_{\rm obs} = hc/E_{\rm TeV}$) and inverting the inequality gives a lower bound on α parallel to Eq. (34),

$$\alpha \;\gtrsim\; \frac{\ln\!\big(2\pi D_C/\lambda_{\rm obs}(E_{\rm TeV})\big) - \tfrac{1}{2}\ln\sigma^2_{\phi,\max}}{\ln(D_C/\ell_P)} \;\approx\; \alpha_{\rm TeV}. \tag{30}$$

The numerator is dominated by the first term, which grows logarithmically with the carrier wavenumber $2\pi/\lambda_{\rm obs}$ and is therefore far larger at TeV than at the MeV line energies of the secondary probe; the $\sigma^2_{\phi,\max}$ term enters only as its half-logarithm and is nearly inert. Evaluating at a representative $E_{\rm TeV} \sim 1$ TeV gives $\alpha_{\rm TeV} \approx 0.7$.[VERIFY] This bound is stronger than the line-width bound of Section V A 5 below and is logically independent of it — it uses the absolute flux rather than the relative line shape — while resting on the identical $\sigma_\phi^2 \propto \nu^2$ suppression physics that underlies the CMB constraint of Section V B.

Insert LHAASO photon count N and EBL-corrected expected count N_0; do not invent these. Confirm $\sigma^2_{\phi,\max} = \ln(N_0/N) = \mathcal{O}(\text{few})$. Recompute $\alpha_{\rm TeV}$ from Eq. (30) using the verified $\sigma^2_{\phi,\max}$ and the representative LHAASO photon energy $E_{\rm TeV}$ (expected $\alpha \gtrsim 0.66$–0.73); flag the chosen $E_{\rm TeV}$.

4. The loophole, and why it is double-edged

The TeV bound can be evaded in exactly one way: if the pedestal kernel $\hat{R}$ is light-tailed rather than Lorentzian, so that a non-negligible fraction of the redistributed flux refills the observing band and partially replenishes the suppressed core. Quantifying such a tail would require resolving the foam on scales $\ell_P \ll \lambda$ — a genuinely sub-wavelength treatment of the metric fluctuations, beyond the eikonal (phase-only) reduction used here and throughout the literature. But this escape is double-edged. The same eikonal derivation that fixes the pedestal shape also produces the suppression amplitude $e^{-\sigma_\phi^2}$ itself; the two are not separable outputs. To disown the Lorentzian pedestal tail at TeV — in order to refill the band — is to disown the very $e^{-\sigma_\phi^2}$, $\sigma_\phi^2 \propto \nu^2$ machinery on which the CMB bound

(Section V B) and the line-width bound (Section V A 5) equally rest. The loophole, if invoked, does not merely soften the TeV bound: it invalidates the entire framework's predictive content at once. A self-consistent foam phenomenology must therefore accept the TeV bound or abandon the suppression physics altogether.

5. Secondary cross-check: energy-dependent line width

The line-width probe is complementary precisely because it is *relative*. The absolute suppression $e^{-\sigma_\phi^2(\nu)}$ attenuates line peak and adjacent continuum identically and so cancels in the fractional width σ/E; the line-width bound is therefore *pedestal-independent*, immune to the redistribution physics that the loophole above contests. That immunity is its virtue. Its vice is that it bites only in the deep strong-foam regime $\sigma_\Delta \gtrsim \tau_G$ and yields a significantly weaker constraint. It rests on the same dynamical-foam Assumption (i) as the TeV bound — the two are consistent siblings, not independent assumptions.

Two independent analyses report a narrow gamma-ray emission feature in the Fermi/GBM data, interpreted as blueshifted electron–positron pair annihilation [11, 18], at observed energies spanning 6–37 MeV; the foam analysis depends only on the Doppler-invariance of the line's fractional width, not on the emission mechanism. Zhang et al. [18] detect a Gaussian line whose centroid evolves as $E_{\rm line} \propto t^{-1}$ from $\approx 37\,$MeV to $\approx 6\,$MeV, with a fractional width $r_{\rm obs} \equiv \sigma_{\rm obs}/E_{\rm line} = 0.10 \pm 0.01$ constant across the seven well-constrained sub-intervals spanning the 9–37 MeV lever arm ($\chi^2/{\rm dof} = 4/7$). In the comoving frame the line has fixed fractional width $r' = \sigma'/E'$; under a bulk Doppler boost $\mathcal{D}$ both centroid and width scale by $\mathcal{D}$, so $r_{\rm int} \equiv \sigma_{\rm int}/E = r'$ is Lorentz-invariant and epoch-independent. The Gaussian-source spectrum Eq. (23) then gives the observed fractional width as a multiplicative excess over $r_{\rm int}$,

$$r_{\rm obs}^2(E) \;=\; r_{\rm int}^2\big[1 + \sigma_\Delta^2/\tau_G^2(E)\big] \;=\; r_{\rm int}^2 + r_{\rm foam}^2(E), \tag{31}$$

with $r_{\rm foam}^2(E) \equiv r_{\rm int}^2\,\sigma_\Delta^2/\tau_G^2(E)$. In the dynamical limit $\sigma_\Delta = \sqrt{2}\,\sigma_\ell/c$ saturates, and with $\tau_G(E) = \lambda_{\rm obs}(E)/(2\pi c\, r_{\rm int})$ the foam excess becomes

$$r_{\rm foam}(E) \;=\; \sqrt{2}\, r_{\rm int}^2\, \sigma_\phi(E) \;=\; r_{\rm int}^2 \cdot \frac{2\sqrt{2}\,\pi\,\sigma_\ell}{\lambda_{\rm obs}(E)} \;\propto\; E, \tag{32}$$

so a χ^2 fit of the seven measurements to the two-parameter model

$$r_{\rm obs}^2(E) \;=\; r_{\rm int}^2 + r_{\rm foam,0}^2\,(E/E_0)^2, \qquad E_0 = 37\,{\rm MeV}, \tag{33}$$

constrains the foam amplitude. The best fit lies at the boundary $r_{\rm foam,0} = 0$ ($r_{\rm int} = 0.111$, $\chi^2_{\rm min} = 5.3$), confirming the constant-$r_{\rm obs}$ hypothesis; profiling over $r_{\rm int}$ gives the one-sided 95% CL upper limit $r_{\rm foam,0}^{\rm max} = 0.27$ at $\Delta\chi^2 = 2.71$ (the boundary-corrected Chernoff threshold [22]). Converting via Eq. (32) and $\sigma_\ell^2 = D_C^{2(1-\alpha)}\ell_P^{2\alpha}$ (Eq. (1), $A_\alpha = 1$) gives

$$\alpha \;\gtrsim\; \frac{\ln(D_C/\lambda_{\rm obs}) + |\ln r_{\rm foam,0}^{\rm max}| - |\ln(2\sqrt{2}\,\pi\, r_{\rm int}^2)|}{\ln(D_C/\ell_P)} \;=\; \frac{89.3 + 1.3 - 3.3}{138.4} \;\approx\; 0.63\,, \tag{34}$$

weaker than the TeV bound of Eq. (30) as anticipated. At this limit the ratio $\sigma_\Delta/\tau_G(E_0) \approx 4.1$ places the bound deep in the strong-foam regime $\sigma_\Delta \gtrsim \tau_G$, where Eq. (21) predicts not only the constrained line-width broadening but also a line-center redshift $\nu_0^{\rm eff}/\nu_0 = \tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$. This is the probe's chief internal caveat: the lever-arm fit uses observed line centers as if intrinsic, yet at the upper limit the foam-induced centroid factor would run from $\nu_0^{\rm eff}/\nu_0 \approx 0.06$ at 37 MeV to ≈ 0.7 at 6 MeV, a strongly energy-dependent depression in mild tension with the smooth $E_{\rm line} \propto t^{-1}$ Doppler-deceleration trend in the data. A joint center-plus-width fit is required for self-consistency at $\alpha \approx 0.63$ and would likely tighten the bound; the present width-only result should be read as the conservative, pedestal-independent floor beneath the primary TeV constraint.

B. CMB

For a broadband thermal source such as the CMB, foam does not introduce a spectral-line shift but instead redistributes power: the coherent-core component is suppressed by $e^{-\sigma_\phi^2(\nu)}$ and the missing flux reappears in a broad pedestal whose shape depends on the short-time behavior of the correlation function $C_\alpha(\tau)$. Because $\sigma_\phi^2 \propto \nu^2$ (Eq. (8)), the foam deficit is concentrated at high frequencies while the pedestal fills in at low frequencies, producing a spectral signature distinct from standard chemical-potential (μ) and Compton-y distortions.

The Far-InfraRed Absolute Spectrophotometer (FIRAS) aboard the Cosmic Background Explorer (COBE) measured the absolute spectrum of the CMB monopole over 60–600 GHz by direct comparison against an internal cryogenic blackbody calibrator — the absolute-photometry feature that distinguishes it from relative-photometry experiments and makes it well suited to constraining spectral *shape* distortions. The calibrated monopole agrees with a pure blackbody to within $\sim 50\,\mathrm{ppm}$ (a few $\times 10^{-5}$ fractional) across the band, placing the still-standard limits $|\mu| \lesssim 9 \times 10^{-5}$ and $|y| \lesssim 1.5 \times 10^{-5}$ [23]. These limits and the 43-channel calibrated spectrum form the empirical basis for the foam constraint derived in this subsection, which sets up the broadband signal, derives it from first principles, and analyzes the FIRAS sensitivity.

1. Foam phase variance at CMB frequencies

CMB photons travel from the last-scattering surface at comoving distance $D_{\mathrm{C}} = 13.87\,\mathrm{Gpc}$ (Planck 2018 flat-ΛCDM cosmology [20]; $H_0 = 67.36\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$). The foam phase variance follows Eq. (8) at the observed wavelength λ_{obs}. The effective foam amplitude $A_{\mathrm{eff}} = A_\alpha D_{\mathrm{C}}^{2(1-\alpha)} \ell_P^{2\alpha}$ used in Section V B 3 enters as

$$\sigma_\phi^2(\nu) = \left(\frac{2\pi\nu}{c}\right)^2 A_{\mathrm{eff}}, \tag{35}$$

with $A_\alpha = 1$.

At $\nu = 150\,\mathrm{GHz}$ this gives

$$\sigma_\phi^2(150\,\mathrm{GHz}) \approx \begin{cases} 7 \times 10^{-2} & \alpha = 1/2 \text{ (random walk)}, \\ 2 \times 10^{-22} & \alpha = 2/3 \text{ (holographic)}. \end{cases} \tag{36}$$

The strong ν^2 dependence concentrates foam effects at high frequency; for $\alpha = 1/2$ the variance approaches unity at the top of the FIRAS band ($\sigma_\phi^2 \approx 1.2$ at 639 GHz), and the linearised template holds for $\alpha \gtrsim 0.50$, where $\sigma_\phi^2(\nu) \lesssim 1$ through the band (Section V B 3).

2. Foam-modified CMB spectrum

The broadband blackbody admits a direct evaluation: compute $g_{\mathrm{src}}^{(1)}(\tau)$ as the Fourier transform of $B_\nu(T_0)$ via Eq. (22), apply the jitter-convolution Eq. (16), and inverse-transform to recover $S_{\mathrm{obs}}(\nu)$. This route entangles the model-independent per-mode suppression $e^{-\sigma_\phi^2(\nu)}$ with the C_α-dependent redistribution of the suppressed flux. We instead present a derivation that isolates these two contributions: the coherent core, which depends only on the asymptotic phase variance $\sigma_\phi^2(\nu)$ and is therefore obtained in closed form independent of foam temporal dynamics, and the pedestal kernel $\hat{R}(\nu - \nu'; \nu')$, which carries the full dependence on the short-τ behavior of $C_\alpha(\tau)$ and is treated phenomenologically pending a first-principles specification of the foam temporal autocorrelation. A worked example with exponential $C_\alpha(\tau)$ below makes the pedestal kernel concrete and identifies the physically-realized case used in the Fisher analysis of Section V B 3.

Decomposing the blackbody as an incoherent superposition of narrowband modes $S_{\mathrm{src}}(\nu') = B_{\nu'}(T_0)$, each mode is modulated by the *same* achromatic jitter $\delta t(t)$, with per-mode foam suppression

$$\mathcal{F}_{\mathrm{foam}}(\tau; \nu') = e^{-\sigma_\phi^2(\nu')[1 - C_\alpha(\tau)]}, \quad \sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 \sigma_\ell^2(D), \tag{37}$$

that approaches the constant $e^{-\sigma_\phi^2(\nu')}$ for all $\tau \neq 0$ in the $C_\alpha \to 0$ limit. For the CMB line of sight, $\sigma_\ell^2(D)$ here is the comoving cumulative variance of Section V B 1, so that $\sigma_\phi^2(\nu') = (2\pi\nu'/c)^2 A_{\mathrm{eff}}$, the cumulative comoving foam variance used in the fit.

For a single narrowband mode at frequency ν', the source coherence is $g_{\mathrm{src}}^{(1)}(\tau) = e^{-i2\pi\nu'\tau}$ (the $\tau_G \to \infty$ limit of Eq. (19)), and the jitter convolution gives

$$g_{\mathrm{obs}}^{(1)}(\tau) = e^{-i2\pi\nu'\tau}\, \mathcal{F}_{\mathrm{foam}}(\tau; \nu') = e^{-i2\pi\nu'\tau}\, e^{-\sigma_\phi^2(\nu')[1 - C_\alpha(\tau)]}.$$

In the dynamical limit, $\mathcal{F}_{\mathrm{foam}}(\tau; \nu')$ transitions from $\mathcal{F}_{\mathrm{foam}}(0; \nu') = 1$ to the plateau $e^{-\sigma_\phi^2(\nu')}$ within the narrow window $|\tau| \lesssim \tau_{\mathrm{foam}}$. The Fourier transform of the plateau (constant in τ outside that window) is $e^{-\sigma_\phi^2(\nu')}\delta(\nu - \nu')$ — the

coherent core at this mode's frequency — while the Fourier transform of the rise from plateau to unity is supported on $|\tau| \lesssim \tau_{\rm foam}$ and is therefore broad in frequency, of width $\sim 1/\tau_{\rm foam}$ — the pedestal at this mode. The split applied below to $\mathcal{F}_{\rm foam}(\tau;\nu')$ generalizes this single-mode decomposition to the full Planck spectrum by linear superposition.

To isolate the coherent-core and pedestal contributions, we split $\mathcal{F}_{\rm foam}(\tau;\nu') = e^{-\sigma_\phi^2(\nu')} + [\mathcal{F}_{\rm foam}(\tau;\nu') - e^{-\sigma_\phi^2(\nu')}]$: the first term contributes the constant asymptote at all lags (Fourier-transforming to a delta function at $\nu = \nu'$), while the residual is localized at $|\tau| \lesssim \tau_{\rm foam}$ (Fourier-transforming to a broad pedestal). Taking the intensity-weighted Fourier transform via Eq. (22) yields

$$S_{\rm obs}(\nu) = e^{-\sigma_\phi^2(\nu)}\, B_\nu(T_0) + \int_0^\infty B_{\nu'}(T_0)\, \hat{R}(\nu-\nu';\nu')\, d\nu', \tag{38}$$

where $\hat{R}$ denotes the residual's Fourier transform. The coherent-core term $e^{-\sigma_\phi^2(\nu)} B_\nu(T_0)$ is exact and model-independent (requiring only achromatic jitter and $C_\alpha \to 0$); the pedestal kernel $\hat{R}$ depends on the short-τ behavior of $C_\alpha(\tau)$ — the foam coherence timescale $\tau_{\rm foam}$ — and is therefore undetermined without specifying C_α. This is the broadband generalization of the core-plus-pedestal decomposition stated in Section IV. Because the per-mode phase modulation $e^{i\omega\delta t(t)}$ is unitary, any physical pedestal kernel $\hat{R}$ derived from a specific $C_\alpha(\tau)$ conserves photon number per mode: $\int [S_{\rm obs}(\nu) - B_\nu(T_0)]\, d\nu/(h\nu) = 0$.

To make the pedestal kernel concrete, we evaluate $\hat{R}$ for the minimal one-parameter model $C_\alpha(\tau) = e^{-|\tau|/\tau_{\rm foam}}$, the autocorrelation of a Markovian metric fluctuation. The framework does not require this specific choice; any $C_\alpha(\tau)$ with characteristic decorrelation time $\tau_{\rm foam} \ll 1/\nu_{\rm max}$ (satisfied with enormous margin for Planck-scale foam, $\tau_{\rm foam} \sim \ell_P/c \approx 5\times10^{-44}\,$s) has a Fourier transform of width $\gtrsim 1/\tau_{\rm foam} \gg \nu_{\rm max}$, so the pedestal kernel is constant across the FIRAS band irrespective of its functional form, and the conclusion that the in-band pedestal is negligible follows from the bandwidth ratio alone. Substituting into Eq. (12) gives

$$\sigma_\Delta^2(\tau) = \frac{2\sigma_\ell^2(D)}{c^2}\big[1 - e^{-|\tau|/\tau_{\rm foam}}\big]. \tag{39}$$

The Fisher analysis below operates in the linearised regime $\sigma_\phi^2(\nu') \ll 1$. Expanding $\mathcal{F}_{\rm foam}(\tau;\nu')$ to first order and subtracting the long-lag asymptote $e^{-\sigma_\phi^2(\nu')}$ leaves the short-τ residual

$$\mathcal{F}_{\rm foam}(\tau;\nu') - e^{-\sigma_\phi^2(\nu')} \;\approx\; \sigma_\phi^2(\nu')\, e^{-|\tau|/\tau_{\rm foam}}, \tag{40}$$

whose Fourier transform is a Lorentzian centered on the source frequency,

$$\hat{R}(\nu-\nu';\nu') \;=\; \sigma_\phi^2(\nu')\, \frac{2\tau_{\rm foam}}{1 + [2\pi(\nu-\nu')\tau_{\rm foam}]^2}, \tag{41}$$

of full width at half maximum $1/(\pi\tau_{\rm foam})$. For Planck-scale foam $\tau_{\rm foam} \sim \ell_P/c \approx 5\times10^{-44}\,$s the Lorentzian width is $\sim 6\times10^{42}\,$Hz, eleven orders of magnitude beyond the FIRAS upper edge $\nu_{\rm max} = 6\times10^{11}\,$Hz; the fraction of the redistributed flux falling within the FIRAS band is $\sim 10^{-31}$. The physical pedestal therefore deposits no detectable in-band power for any plausible $\tau_{\rm foam}$. The constraint of Section V B 3 consequently adopts the no-pedestal limit as the primary template.

To probe robustness against in-band pedestal contamination beyond the physical-kernel result above, we also evaluate the Fisher matrix with a *μ-aligned* band-filling redistribution kernel, deliberately shaped $\propto 1/\nu$ to match the Rayleigh–Jeans form of the μ-distortion template (Section V B 3) and thereby maximize the foam–μ Fisher degeneracy — the most effective way an in-band pedestal could disguise foam as a standard μ-distortion:

$$K_\nu^{\rm fast}(\nu) \;=\; \begin{cases} \dfrac{1}{\nu\,\ln(\nu_{\rm max}/\nu_{\rm min})} & \nu_{\rm min} \le \nu \le \nu_{\rm max}, \\ 0 & \text{otherwise}, \end{cases} \tag{42}$$

with $\nu_{\rm min} = 60\,$GHz and $\nu_{\rm max} = 600\,$GHz set by the FIRAS band edges. $K_\nu^{\rm fast}(\nu)$ is a normalized density in observer frequency ($\int_{\nu_{\rm min}}^{\nu_{\rm max}} K_\nu^{\rm fast}\, d\nu = 1$, units Hz^{-1}) that specifies the fraction of the total foam-removed flux reappearing as pedestal intensity per Hz at observer frequency ν; the form above corresponds to uniform energy redistribution in $\ln\nu$ across the band. Within the FIRAS band, $K_\nu^{\rm fast}(\nu)$ replaces the physical pedestal contribution to Eq. (38) by a μ-aligned factorised substitution

$$\int B_{\nu'}(T_0)\, \hat{R}(\nu-\nu';\nu')\, d\nu' \;\longrightarrow\; \sigma_\ell^2(D)\, K_\nu^{\rm fast}(\nu)\, P_{\rm foam}, \quad P_{\rm foam} \equiv \int_{\nu_{\rm min}}^{\nu_{\rm max}} \left(\frac{2\pi\nu'}{c}\right)^2 B_{\nu'}(T_0)\, d\nu', \tag{43}$$

with $P_{\rm foam}$ the total foam-removed flux per unit σ_ℓ^2. The substitution forces the pedestal kernel to factorise into a ν-only template ($K_\nu^{\rm fast}$) times a ν'-only weighting, a structure no physical $\hat{R}(\nu-\nu';\nu')$ satisfies (the Lorentzian above carries ν'-dependence through $\sigma_\phi^2(\nu')$). The template is energy-conserving but not photon-number-conserving, and serves purely as a μ-aligned probe of the foam–μ kernel-shape degeneracy. Its effect on the bound is evaluated in Section V B 4.

3. Constraint from FIRAS

We use $I_\nu \equiv S_{\rm obs}(\nu)$ to denote the observed CMB intensity including the foam signal. Working in the linearised regime $\sigma_\phi^2(\nu) \ll 1$, we treat four spectral parameters $\boldsymbol{\theta} = (A_{\rm eff}, T_0, \mu, y)$, where $A_{\rm eff} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ is the cumulative comoving foam variance at the CMB distance (Section V B 1), so that $\sigma_\phi^2(\nu) = (2\pi\nu/c)^2 A_{\rm eff}$. Note that α is not fitted directly; only the lumped amplitude $A_{\rm eff}$ enters the fit, and bounds on α are obtained *a posteriori* by inverting the Ng–Perlman relation at fixed $A_\alpha = 1$.

In the physically-realized regime (Section V B 2) the redistributed pedestal lies at $\sim 10^{42}$ Hz, eleven orders of magnitude above the FIRAS band, so the only in-band foam signature is the pure deficit

$$\frac{\partial I_\nu}{\partial A_{\rm eff}} = -\left(\frac{2\pi\nu}{c}\right)^2 B_\nu(T_0), \tag{44}$$

a ν^2-weighted (blue-tilted) Planck spectrum. The chemical-potential and Compton-y distortion templates are

$$\frac{\partial I_\nu}{\partial \mu} = -B_\nu(T_0)\,\frac{x e^x}{(e^x-1)^2}, \tag{45}$$

$$\frac{\partial I_\nu}{\partial y} = \quad B_\nu(T_0)\,\frac{x e^x}{e^x-1}\left[\frac{x(e^x+1)}{e^x-1} - 4\right], \tag{46}$$

with $x = h\nu/(k_B T_0)$, encoding respectively early-universe energy injection (Silk damping, decaying particles) and late-time electron heating (reionization, cluster gas).

We constrain $A_{\rm eff}$ by fitting these templates — together with the FIRAS Galactic-foreground template — to the public COBE/FIRAS monopole residual spectrum and its calibration covariance [23]. Raising T_0 brightens the whole Planck curve, whereas the foam deficit suppresses it with an extra ν^2 weighting; over the CMB-bright band (the peak through the onset of the Wien fall-off) the two differ only by a spectral tilt, so the foam template is nearly indistinguishable from a temperature shift ($\rho_{A_{\rm eff} T_0} \simeq 1$ in the fit) and, through that near-degeneracy, strongly collinear with the y and μ templates (quantified by the forecast correlation matrix Eq. (48) below). Using the full 43×43 apodization covariance, the generalized least-squares fit over $(T_0, \mu, y, A_{\rm eff})$ and the foreground amplitude recovers μ and y consistent with zero and with the Fixsen limits ($|\mu| < 1.7 \times 10^{-4}$, $|y| < 1.0 \times 10^{-5}$ at 95%; $\chi^2/\nu = 1.24$) and a foam amplitude consistent with zero,

$$A_{\rm eff} = (-2.4 \pm 5.3) \times 10^{-11}\,{\rm m}^2. \tag{47}$$

Inverting $A_{\rm eff} = A_\alpha D_C^{2(1-\alpha)} \ell_P^{2\alpha}$ at $A_\alpha = 1$, the one-sided 95% upper limit $A_{\rm eff} < 1.645\,\sigma(A_{\rm eff}) = 8.7 \times 10^{-11}\,{\rm m}^2$ gives the lower bound $\alpha \gtrsim 0.52$ (collected with the forecasts below in Table I). The one direction that could still break the foam–T_0 degeneracy is deep into the Wien/sub-mm regime, where the ν^2-tilt finally separates from a temperature shift, but galactic dust and the cosmic infrared background dominate — beyond the FIRAS band, and their foreground confusion makes the monopole impractical for breaking the degeneracy there. The linearised template $\Delta I_\nu \approx \sigma_\phi^2(\nu)\,[\text{template}]$, valid for $\sigma_\phi^2 \ll 1$, holds across the FIRAS band at the headline threshold (σ_ϕ^2 exceeds unity only for stronger foam, $\alpha \lesssim 0.50$; Section V B 1): replacing the deficit $-\sigma_\phi^2 B_\nu$ by the saturated form $-(1-e^{-\sigma_\phi^2})B_\nu$ changes the integrated in-band amplitude by $\approx 13\%$ at $\alpha = 0.50$ but by $< 1\%$ at $\alpha \gtrsim 0.52$ ($\Delta\alpha \sim 3 \times 10^{-5}$), so the bound is unaffected; nonlinearity matters only for stronger foam ($\alpha \lesssim 0.50$), where it may also help break the foam–μ degeneracy.

That same near-degeneracy has two consequences for the CMB temperature itself. First, admitting foam as a free parameter inflates the marginalized temperature uncertainty: the FIRAS spectral-shape determination of T_0 degrades from 1.2×10^{-5} K to 7.7×10^{-4} K (a factor ~ 64; $\sigma(y)$ and $\sigma(\mu)$ inflate by ~ 17 and ~ 3), the latter comparable to the full COBE/FIRAS calibration uncertainty 5.7×10^{-4} K [24] — so a foam-agnostic monopole analysis roughly doubles the temperature error budget, and the quoted precision of $T_0 = 2.72548$ K implicitly assumes no foam. Second, foam present but *omitted* from the fit shifts the central value by $\Delta T_0 = b_{T_0} A_{\rm eff}$ with $b_{T_0} \approx -1.5 \times 10^7\,{\rm K\,m^{-2}}$; for $A_\alpha = 1$

this is only $\approx 3\times10^{-18}\,\mathrm{K}$ at the holographic $\alpha = 2/3$ and vanishes as $\alpha \to 1$ (where the comoving foam variance ceases to grow with distance), reaching $\sigma(T_0)$ only at $\alpha \approx 0.52$ where the foam is already a many-σ signal. Both effects therefore matter only for foam near FIRAS detectability; the recovered T_0 is immune to any foam consistent with the GRB bound $\alpha \gtrsim 0.63$.

4. *Fisher forecasts for future spectrometers*

To forecast the reach of future absolute spectrometers we propagate the same no-pedestal templates through a Fisher matrix. Calibrated to FIRAS with a flat per-channel noise $\sigma_\nu = 0.01\,\mathrm{MJy/sr}$ (a band-averaged stand-in for the measured errors), the Fisher forecast gives $\sigma(A_{\rm eff}) = 2.2\times10^{-11}\,\mathrm{m}^2$ and the same $\alpha \gtrsim 0.52$, reproducing the data fit; the conditional value $\sigma(A_{\rm eff})|_{\mu,y\,\mathrm{fixed}} = 8.2\times10^{-13}\,\mathrm{m}^2$ shows that marginalizing over the distortions costs a factor ~ 27 through the foam–T_0 collinearity. The fiducial degeneracy structure is captured by the correlation matrix for $(A_{\rm eff},\, T_0,\, \mu,\, y)$,

$$\rho_{ij}^{\rm fid} = \begin{pmatrix} 1.000 & \mathbf{0.9997} & 0.928 & 0.999 \\ \mathbf{0.9997} & 1.000 & 0.934 & 0.998 \\ 0.928 & 0.934 & 1.000 & 0.915 \\ 0.999 & 0.998 & 0.915 & 1.000 \end{pmatrix}, \tag{48}$$

in which the foam deficit is almost perfectly aligned with a temperature shift ($\rho_{A_{\rm eff}T_0} = 0.9997$, bold) and strongly collinear with the y and μ distortions (0.999 and 0.928). Defining the marginal detection SNR as $A_{\rm eff}/\sigma(A_{\rm eff})$, so that SNR = 1.645 marks the one-sided 95% CL threshold, Figure 1 plots it versus α, and Table I collects the thresholds.

TABLE I. Marginal foam-detection thresholds (one-sided 95% CL): *lower* bounds on the exponent α ($A_\alpha = 1$), since the foam signal $\sigma_\ell^2 \propto (\ell_P/D_C)^{2\alpha}$ decreases with α and a non-detection therefore excludes small α. FIRAS is from the data fit (Section V B 3); PIXIE, SuperPIXIE, and Voyage 2050 from the Fisher forecast scaled by their published μ-sensitivities [25]. The three forecast entries (0.545, 0.549, 0.554) round to 0.54–0.55; the thresholds increase monotonically.

	lower bound
FIRAS	$\alpha \gtrsim 0.52$
PIXIE	$\alpha \gtrsim 0.54$
SuperPIXIE	$\alpha \gtrsim 0.55$
Voyage 2050	$\alpha \gtrsim 0.55$

To bound the impact of μ-aligned in-band pedestal contamination, we repeat the analysis with the μ-aligned band-filling kernel of Section V B 2, replacing the deficit by

$$\frac{\partial I_\nu}{\partial A_{\rm eff}} = -\left(\frac{2\pi\nu}{c}\right)^2 B_\nu(T_0) + K_\nu^{\rm fast}(\nu)\int_{\nu_{\rm min}}^{\nu_{\rm max}} \left(\frac{2\pi\nu'}{c}\right)^2 B_{\nu'}(T_0)\,d\nu', \tag{49}$$

whose $1/\nu$ pedestal we deliberately gave the Rayleigh–Jeans shape of the μ-distortion template (Eq. (42); the Rayleigh–Jeans limit of Eq. (45) is $\Delta I/B \to -\mu/x \propto \nu^{-1}$) so as to maximize the foam–μ overlap. The resulting correlation matrix for $(A_{\rm eff},\, T_0,\, \mu,\, y)$ is

$$\rho_{ij}^{\mu\text{-align}} = \begin{pmatrix} 1.000 & 0.937 & \mathbf{0.976} & 0.010 \\ 0.937 & 1.000 & 0.969 & -0.212 \\ \mathbf{0.976} & 0.969 & 1.000 & -0.127 \\ 0.010 & -0.212 & -0.127 & 1.000 \end{pmatrix}. \tag{50}$$

Compared with the fiducial matrix Eq. (48), adding the in-band pedestal rotates the foam template off the temperature axis ($\rho_{A_{\rm eff}T_0}$: $0.9997 \to 0.937$) and off the y template ($\rho_{A_{\rm eff}y}$: $0.999 \to 0.010$), leaving the engineered foam–μ degeneracy ($\rho_{A_{\rm eff}\mu} = 0.976$, bold) as the dominant one. Because 0.976 is milder than the 0.9997 it replaces, the amplitude is in fact *better* constrained,

$$\sigma(A_{\rm eff}) = 1.7\times10^{-12}\,\mathrm{m}^2, \qquad \sigma(A_{\rm eff})\big|_{\mu,y\,\mathrm{fixed}} = 2.9\times10^{-13}\,\mathrm{m}^2, \tag{51}$$

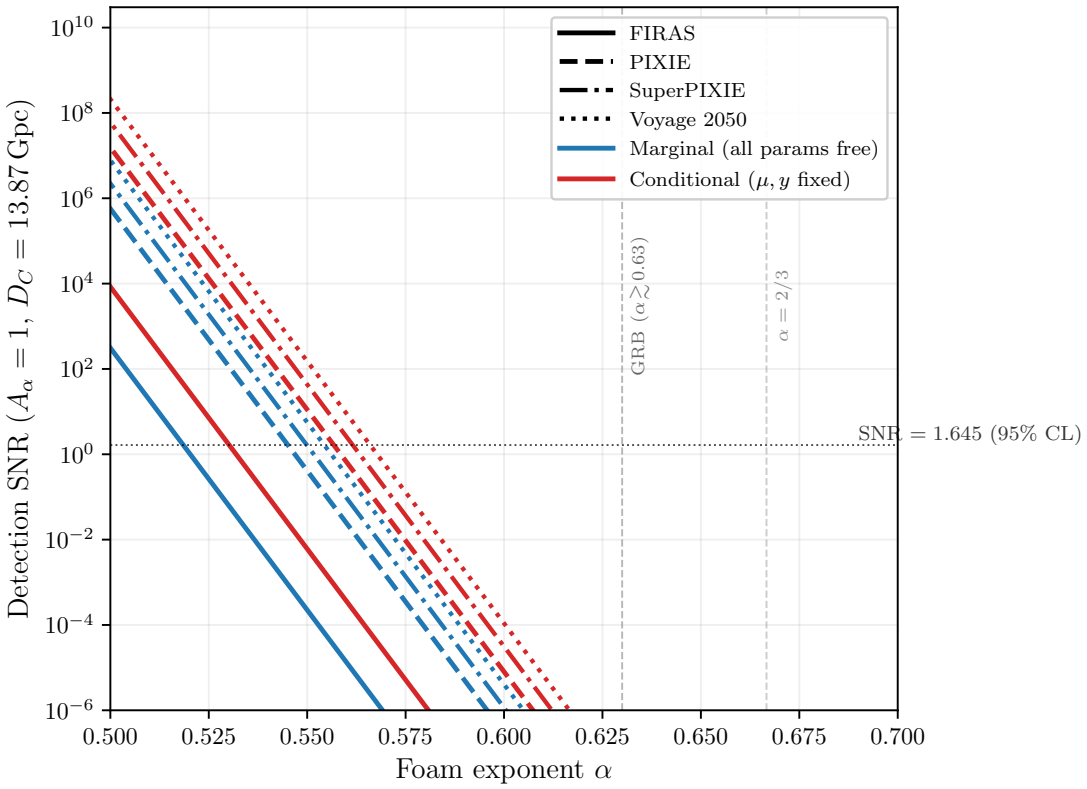


FIG. 1. Detection SNR for the foam amplitude $A_{\rm eff}$ as a function of foam exponent α ($A_\alpha = 1$, $D_C = 13.87\,$Gpc). Blue curves: marginal SNR (all four parameters $A_{\rm eff}, T_0, \mu, y$ free). Red curves: conditional SNR (μ, y held fixed), which is $\sim 27\times$ higher. Line styles denote the instrument: solid FIRAS, dashed PIXIE, dash-dotted SuperPIXIE, dotted Voyage 2050, with forecast μ-sensitivities scaled relative to FIRAS [25]. The marginal lower bounds (SNR = 1.645, one-sided 95% CL) cluster at $\alpha \approx 0.52$–0.55, illustrating the logarithmic saturation of the bound with sensitivity. The dotted horizontal line marks SNR = 1.645 (one-sided 95% CL); vertical dashed lines mark the spectroscopic lower bound from GRB 221009A ($\alpha \gtrsim 0.63$) and the holographic prediction $\alpha = 2/3$.

a factor ~ 6 marginalization penalty driven by the foam–μ collinearity, and the threshold *tightens* slightly to $\alpha \gtrsim 0.53$. The μ-aligned in-band pedestal therefore does not loosen the bound: the factor $N \approx 13$ between the two amplitude uncertainties is compressed by the logarithmic Ng–Perlman relation $\sigma_\ell^2 = D_C^{2(1-\alpha)}\ell_P^{2\alpha}$ into a negligible shift $\Delta\alpha = \ln N/[2\ln(D_C/\ell_P)] \approx 0.009$. An external constraint on μ helps most in this μ-limited regime: imposing the ΛCDM dissipation prediction $\mu \approx 2\times10^{-8}$ [25] as a prior (five orders of magnitude tighter than the FIRAS μ measurement, hence effectively fixing μ) sharpens $\sigma(A_{\rm eff})$ by $\sim$ 3–5×, though the logarithmic relation again limits the threshold gain to $\Delta\alpha \lesssim 0.005$; in the physical no-pedestal case, where the limiting degeneracy is with T_0 rather than μ, even fixing μ barely helps. Either way, the worked example of Section V B 2 shows that any exponential-class C_α with Planck-scale $\tau_{\rm foam}$ lies in the no-pedestal regime, so the physical $\alpha \gtrsim 0.52$ is the headline and the μ-aligned $\alpha \gtrsim 0.53$ merely confirms its robustness.

The proposed PIXIE satellite [26] — a fully cryogenic ($T_{\rm sys} \approx 2.7\,$K) nulling polarizing Fourier transform spectrometer spanning 30 GHz to 6 THz — is projected to surpass FIRAS sensitivity by roughly three orders of magnitude, reaching $|\mu| \lesssim 10^{-8}$ and $|y| \lesssim 10^{-9}$ and entering the regime of guaranteed ΛCDM spectral signals from Silk damping and reionization. Propagating this gain through the Fisher-matrix scaling derived above, the marginal foam detection threshold shifts from $\alpha \gtrsim 0.52$ (FIRAS) to approximately $\alpha \gtrsim 0.54$ (PIXIE; Table I), narrowing the gap with the spectroscopic lower bound $\alpha \gtrsim 0.63$ from GRB 221009A (Fig. 1). All thresholds are quoted for the physical no-pedestal deficit. The step sizes track the logarithm of the sensitivity ratio — each decade of $\sigma(\mu)$ buys only $\Delta\alpha = \ln 10/[2\ln(D_C/\ell_P)] \approx 0.008$ — so the $\sim$3-decade FIRAS→PIXIE gain accounts for the ≈ 0.03 step, whereas PIXIE→Voyage 2050 (under a decade combined) adds only ≈ 0.009; the thresholds saturate monotonically (the three forecast entries round 0.545, 0.549, 0.554 to 0.54–0.55). The limiting degeneracy — with a temperature shift for the no-pedestal deficit — persists on the linearized template, but PIXIE's extension to 6 THz accesses the nonlinear regime $\sigma_\phi^2 \gg 1$ for $\alpha \lesssim 0.52$ — at higher α than the FIRAS-band crossing $\alpha \lesssim 0.50$ (Section V B 3), since the

wider 6 THz band pushes the $\sigma_\phi^2 = 1$ boundary upward — where the foam deficit diverges markedly from both the blackbody and μ-distortion shapes and the degeneracy may be substantially reduced. The $\alpha \gtrsim 0.54$ projection may therefore understate the reach: if the nonlinear template effectively breaks this degeneracy at PIXIE sensitivity, the true threshold could be significantly more stringent.

Extending the same Fisher analysis to the 10 GHz–3 THz bands of the SuperPIXIE and Voyage 2050 concepts [25] — with forecast distortion sensitivities $\sigma(\mu) \approx 7.7\times10^{-9}$ and 2.2×10^{-9} — shifts the marginal threshold only to $\alpha \gtrsim 0.55$ (Table I), barely beyond the PIXIE projection. The reason is structural rather than instrumental: in the physically realized no-pedestal limit the foam deficit $-\sigma_\phi^2(\nu)B_\nu$ is degenerate with a temperature shift at $\rho^{\rm fid}_{A_{\rm eff}T_0} = 0.9997$, and this degeneracy is *band-robust* — broadening the coverage from the FIRAS band to 10 GHz–6 THz leaves it unchanged to four decimal places. The wider band does break the foam–μ degeneracy probed by the μ-aligned $K_\nu^{\rm fast}$ pedestal (from $\rho^{\mu\text{-align}}_{A_{\rm eff}\mu} = 0.98$ at FIRAS to 0.66 over 10 GHz–3 THz), so future missions are markedly more robust against in-band pedestal contamination, but the physical threshold does not move. Together with the logarithmic Ng–Perlman scaling $\Delta\alpha = \ln N/[2\ln(D_C/\ell_P)]$, which converts even a 10^4-fold sensitivity gain over FIRAS into $\Delta\alpha \lesssim 0.04$, this sets a firm ceiling: as an independent broadband probe — absolute calibrated photometry, no source-profile assumption, and a different systematic-error budget from spectroscopy — spectral-distortion photometry yields an independent lower bound ($\alpha \gtrsim 0.52$ from FIRAS, rising to at most $\alpha \gtrsim 0.55$ for next-generation photometry) that cannot, by sensitivity or bandwidth alone, match the stronger GRB spectroscopic bound $\alpha \gtrsim 0.63$ or reach the holographic value $\alpha = 2/3$. The two channels therefore probe the same exponent through different sources, observables, and systematics — an independent cross-check rather than a joint bracket — with the GRB the more constraining of the two.

VI. CONCLUSIONS

Spacetime foam imprints an achromatic, stochastic arrival-time jitter that modifies the first-order temporal coherence $g^{(1)}(\tau)$, and hence the observed spectrum, providing a spectroscopic probe complementary to the imaging (PSF, Strehl) channel. In the dynamical-foam limit the spectrum splits into a coherent line core suppressed by $e^{-\sigma_\phi^2}$ and an incoherent pedestal, yielding a shape-distortion bound on the Ng–Perlman exponent α. The MeV emission lines of GRB 221009A give $\alpha \gtrsim 0.63$ (one-sided 95% CL), while a fit to the COBE/FIRAS monopole gives $\alpha \gtrsim 0.52$ (95% CL); next-generation spectrometers (PIXIE, SuperPIXIE, Voyage 2050) reach $\alpha \gtrsim 0.54$–0.55, the bound saturating logarithmically because the foam deficit is nearly degenerate with a temperature shift. None of these bounds exclude the holographic value $\alpha = 2/3$.

All single-source spectroscopic bounds derived in this paper assume the dynamical-foam limit $\tau_{\rm foam} \ll \tau_c$ (Section III B), in which the foam path-length jitter decorrelates many times within each photon coherence time. If foam fluctuations are instead slow ($\tau_{\rm foam} \gg \tau_c$), the accumulated phase is essentially frozen during each coherence time and the foam acts as a deterministic, frequency-independent phase screen — no spectral broadening or peak reduction occurs within a single observation, and the shape-distortion bounds do not apply. The CMB constraint (Section V B) is similarly derived in the dynamical limit (Section V B 2 works in $C_\alpha \to 0$): in the opposite quasi-static limit the per-mode foam suppression $\mathcal{F}_{\rm foam}(\tau) \to 1$ (Eq. (18)) and the CMB monopole is unmodified, so the shape-distortion bounds do not apply in either channel.

The dynamical bound $\alpha \gtrsim 0.63$ (Eq. (34)) does not exclude the holographic value $\alpha = 2/3 \approx 0.667$. A principal instrumental limitation of the present analysis is the MeV sensitivity gap: the spectral region between the upper edge of the Fermi-GBM BGO detector response ($\lesssim 40$ MeV, where the effective area falls steeply) and the regime in which Fermi-LAT achieves robust detection efficiency ($\gtrsim 100$ MeV). The frequency $\nu_\star \simeq 37$ MeV at which the foam-induced spectral break is predicted to be most constraining falls squarely within this gap, so neither instrument provides fully resolved coverage across the critical transition. Closing the MeV gap is a recognized community priority, addressed by next-generation satellite concepts that combine Compton-scattering and pair-conversion detection techniques to achieve continuous sensitivity from ~ 100 keV to beyond 1 GeV. The All-sky Medium Energy Gamma-ray Observatory (AMEGO [27]) and its MIDEX-scale variant AMEGO-X [28] are designed to cover 100 keV to 1 GeV with order-of-magnitude improvements over both COMPTEL and Fermi-LAT in the MeV decade; the ESA-proposed M-class concept e-ASTROGAM [29] targets a comparable bandpass. The near-term NASA SMEX mission COSI (scheduled launch ∼2027) will extend broadband sensitivity to 0.2–5 MeV, providing a valuable pathfinder, though its upper energy bound falls well short of the 37 MeV regime and the gap therefore remains open. Realization of any of the broader-bandpass concepts would, for the first time, permit fully resolved spectral measurements of GRB prompt emission across the 37 MeV transition without the instrumental discontinuity that limits the present work; reducing the per-epoch line-width uncertainties to $\delta r \lesssim 0.01$ would tighten the bound to $\alpha \gtrsim 0.64$, removing a major systematic and enabling substantially tighter tests of spacetime-foam models. Independently, a joint fit of the line centroid and width across epochs — the present analysis constrains the width alone — would exploit the energy-dependent line-center shift $\nu_0^{\rm eff} = \nu_0\,\tau_G^2/(\tau_G^2 + \sigma_\Delta^2)$ (Eq. (23)) that the same foam jitter induces, adding a second handle on α and

further tightening the bound.

A complementary implication runs in the opposite direction. The flagship goal of next-generation CMB spectrometers — PIXIE, SuperPIXIE, Voyage 2050 — is a first detection of the chemical potential μ, whose guaranteed ΛCDM value $\mu \simeq 2 \times 10^{-8}$ arises from Silk damping of primordial perturbations. Our Fisher analysis shows that the foam-induced distortion is nearly degenerate with the μ template across the accessible band (Section V B 3); the two cannot be separated from the monopole spectrum alone. A measured μ that *departs* from the ΛCDM prediction would be a landmark signal whatever its origin, and spacetime foam is one propagation-induced alternative to early-universe energy injection — distinguishable only with external information. The GRB lower bound $\alpha \gtrsim 0.63$ already limits any foam contribution to the CMB monopole to far below the ΛCDM μ, so foam does not contaminate the guaranteed-μ target; but this same spectroscopy-plus-GRB complementarity is what would be needed to attribute a larger, anomalous μ. In this sense the CMB μ-distortion measurement and the foam search are two readings of a single spectral degeneracy. The decisive limitation here is theoretical rather than instrumental: the pedestal shape — and hence how completely the foam deficit can be separated from a temperature shift or from μ — is fixed by the foam temporal autocorrelation $C_\alpha(\tau)$, which still awaits a first-principles derivation from a quantum-gravity model. The amplitude A itself is in the same position: it is calibrated phenomenologically rather than derived from a fluctuation-dissipation or holographic-entropy argument at the Planck-cell level.

The multi-epoch $r_{\rm obs}$ dataset constrains any propagation effect that broadens an emission line in proportion to a power of E, not only the Ng–Perlman spacetime-foam template considered here. The most natural extension is to stochastic Lorentz-invariance violation [17], in which the photon group velocity carries a Planck-suppressed random coefficient $v(E) = c\big[1 - \xi\,(E/E_{\rm QG,n})^n\big]$, with ξ of order unity drawn independently for each Planck-scale traversal. Photons in the same line then propagate at slightly different speeds, broadening the observed width as $\sigma_{\rm LIV}(E)/E \propto E^n$ on top of any intrinsic contribution. Unlike the deterministic LIV signal probed by photon time-of-flight studies [30] — where GRB 221009A LHAASO TeV photons already constrain $E_{\rm QG,1} \gtrsim$ a few $\times\, E_{\rm Pl}$ [21] — stochastic LIV produces a *width* signal that survives light-curve cross-correlation, and a multi-epoch $r_{\rm obs}$ measurement is therefore one of the few clean probes available. Fitting Eq. (33) recovers the foam constraint as the $n = 1$ case; refitting the same seven-epoch dataset under $r_{\rm LIV} \propto (E/E_0)^n$ would yield bounds on $\langle\xi^2\rangle^{1/2}/E^n_{\rm QG,n}$ at each order, with the $n = 2$ (quadratic) case particularly valuable because the deterministic time-of-flight tests are weakest there at $z = 0.151$.

ACKNOWLEDGMENTS

This work was supported in part by the Director, Office of Science, Office of High Energy Physics of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231.

The authors used Claude (Anthropic) and Valency Systems to assist with drafting and editing portions of this manuscript, with detailed reference searches, and in developing the Fisher-matrix and GRB multi-epoch analysis scripts. All content, equations, numerical results, and conclusions were reviewed and verified by the authors, who take full responsibility for the work. AI assistance was used in accordance with *Physical Review* editorial policy on computational tools.

We thank Robin Kaiser for helpful discussions that helped motivate this work.

[1] J. A. Wheeler, Physical Review **97**, 511 (1955).
[2] S. W. Hawking, Nuclear Physics B **144**, 349 (1978).
[3] S. Carlip, Rep. Prog. Phys. **86**, 066001 (2023), arXiv:2209.14282 [gr-qc].
[4] Y. J. Ng and E. S. Perlman, Universe **8**, 382 (2022).
[5] Y. J. Ng, Physical Review Letters **86**, 2946 (2001).
[6] Y. J. Ng, Modern Physics Letters A **18**, 1073 (2003).
[7] R. Lieu and L. W. Hillman, The Astrophysical Journal **585**, L77 (2003).
[8] W. A. Christiansen, Y. J. Ng, D. J. E. Floyd, and E. S. Perlman, Physical Review D **83**, 084003 (2011).
[9] E. S. Perlman, S. A. Rappaport, W. A. Christiansen, Y. J. Ng, J. DeVore, and D. Pooley, The Astrophysical Journal **805**, 10 (2015).
[10] V. S. H. Lee, K. M. Zurek, and Y. Chen, Physical Review D **109**, 084005 (2024), arXiv:2312.06757 [gr-qc].
[11] M. E. Ravasio *et al.*, Science **385**, 452 (2024).
[12] E. Steinbring, Galaxies **11**, 115 (2023).
[13] E. Steinbring and Y. J. Ng, arXiv e-prints 10.48550/arXiv.2603.18277 (2026), arXiv:2603.18277.
[14] A. Ishimaru, *Wave Propagation and Scattering in Random Media* (Academic Press, New York, 1978).

[15] S. M. Flatté, R. Dashen, W. H. Munk, K. M. Watson, and F. Zachariasen, *Sound Transmission through a Fluctuating Ocean* (Cambridge University Press, Cambridge, 1979).
[16] B. J. Rickett, Annu. Rev. Astron. Astrophys. **28**, 561 (1990).
[17] G. Amelino-Camelia, Living Reviews in Relativity **16**, 5 (2013), arXiv:0806.0339 [gr-qc].
[18] Y.-Q. Zhang *et al.*, Sci. China Phys. Mech. Astron. (2024), arXiv:2403.12851.
[19] E. Burns *et al.*, Astrophys. J. Lett. **946**, L31 (2023), arXiv:2303.14172.
[20] Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Baccigalupi, *et al.*, Astronomy & Astrophysics **641**, A6 (2020), arXiv:1807.06209.
[21] LHAASO Collaboration, Physical Review Letters **133**, 071501 (2024), arXiv:2402.06009.
[22] H. Chernoff, The Annals of Mathematical Statistics **25**, 573 (1954).
[23] D. J. Fixsen, E. S. Cheng, J. M. Gales, J. C. Mather, R. A. Shafer, and E. L. Wright, Astrophysical Journal **473**, 576 (1996), arXiv:astro-ph/9605054.
[24] D. J. Fixsen, Astrophysical Journal **707**, 916 (2009), arXiv:0911.1955.
[25] J. Chluba *et al.*, Exper. Astron. **51**, 1515 (2021), arXiv:1909.01593.
[26] A. Kogut *et al.*, JCAP **2011** (07), 025, arXiv:1105.2044.
[27] J. McEnery, A. van der Horst, A. Dominguez, A. Moiseev, A. Marcowith, A. Harding, A. Lien, A. Giuliani, A. Inglis, S. Ansoldi, *et al.*, in *Bulletin of the AAS*, Vol. 51 (2019) p. 245, astro2020 Decadal Survey APC White Paper, arXiv:1907.07558 [astro-ph.IM].
[28] R. Caputo, M. Ajello, C. A. Kierans, J. S. Perkins, J. L. Racusin, L. Baldini, E. Burns, P. Cristofari, P. Coppin, P. de la Torre Luque, R. Engel, H. Fleischhack, C. L. Fryer, F. Gargano, A. Giuliani, D. Hartmann, M. Kachelriess, C. Karwin, J. F. Krizmanic, T. R. Lewis, F. Loparco, A. Manfreda, M. N. Mazziotta, J. E. McEnery, S. Mereghetti, J. W. Mitchell, A. A. Moiseev, M. Negro, N. Omodei, E. Orlando, M. Persic, R. Pillera, T. A. Porter, A. Pozanenko, G. Puehlhofer, M. Razzano, C. Sgrò, P. Shawhan, C. Shrader, D. A. Smith, H. Tajima, V. Tatischeff, D. J. Thompson, J. Valverde, T. M. Venters, M. Wood, and G. Zaharijas, Journal of Astronomical Telescopes, Instruments, and Systems **8**, 044003 (2022), arXiv:2208.04990 [astro-ph.IM].
[29] A. De Angelis, V. Tatischeff, M. Tavani, U. Oberlack, I. Grenier, L. Hanlon, R. Walter, A. Argan, P. von Ballmoos, A. Bulgarelli, I. Donnarumma, M. Hernanz, I. Kuvvetli, M. Pearce, A. Zdziarski, J. Conrad, G. Ghisellini, A. Harding, J. Isern, M. Leising, F. Longo, G. Madejski, M. Martinez, M. N. Mazziotta, J. M. Paredes, M. Pohl, R. Rando, M. Razzano, A. Aboudan, M. Ackermann, *et al.*, Experimental Astronomy **44**, 25 (2017), arXiv:1611.02232 [astro-ph.HE].
[30] A. A. Abdo, M. Ackermann, M. Ajello, *et al.* (Fermi GBM/LAT Collaboration), Nature **462**, 331 (2009), arXiv:0908.1832 [astro-ph].

Appendix A: Derivation of the Foam-Modified Coherence

This appendix derives the foam-modified coherence and spectrum directly in the spectral domain for an arbitrary broadband source, recovers the Gaussian-line closed form (21) as the narrow-line limit, and validates the result against a direct Monte-Carlo simulation of the shell model. The foam is a single achromatic path-length fluctuation $\delta t = \delta\ell/c$, so the carrier phase is $\delta\phi = \omega_0\,\delta t$ and carrier and envelope share the one delay δt.

Spectral-domain derivation

Let $E_{\rm src}(t)$ be a general stationary, zero-mean random field,

$$E_{\rm src}(t) = \int d\nu\, \tilde{E}(\nu)\, e^{-i2\pi\nu t}, \qquad \langle \tilde{E}^*(\nu)\tilde{E}(\nu')\rangle = S_{\rm src}(\nu)\,\delta(\nu-\nu'), \tag{A1}$$

the δ-correlation being the Wiener–Khinchin content of stationarity; $g^{(1)}_{\rm src}(\tau) = \int S_{\rm src}(\nu)e^{-i2\pi\nu\tau}d\nu / \int S_{\rm src}(\nu)d\nu$. In the eikonal phase-screen reduction the foam multiplies the component at observed frequency ν by $e^{i\varphi_\nu(t)}$, with $\varphi_\nu(t) = \frac{2\pi\nu}{c}\int (n-1)\,dr = 2\pi\nu\,\delta t(t)$,

$$E_{\rm obs}(t) = \int d\nu\, \tilde{E}(\nu)\, e^{-i2\pi\nu t}\, e^{\,i\varphi_\nu(t)}. \tag{A2}$$

The phase is a sum over the $\sim D/\ell_P$ independent shells, hence Gaussian and zero-mean (CLT), with

$$D_\phi(\nu;\tau) \equiv \left\langle [\varphi_\nu(t+\tau) - \varphi_\nu(t)]^2 \right\rangle = 2\sigma_\phi^2(\nu)\,[1 - C_\alpha(\tau)], \quad \sigma_\phi^2(\nu) = \left(\tfrac{2\pi\nu}{c}\right)^2 \sigma_\ell^2, \tag{A3}$$

the per-frequency phase variance of Eq. (8). The foam is statistically independent of the source. Forming the observed coherence and averaging over source and foam separately, the source δ-correlation collapses $\nu' = \nu$, leaving

TABLE II. Maximum $|B(\tau) - \text{model}|$ over $\tau \in [-3,3]\tau_G$ from a 4×10^6-realization Monte Carlo. CF: closed form (21); FAC: naive factorized form.

	CF	FAC
$\sigma_\Delta/\tau_G = 0.55$	3×10^{-4}	0.12
$\sigma_\Delta/\tau_G = 0.77$	2×10^{-4}	0.18
$\tau_G = 8$ (narrowband)	4×10^{-4}	6×10^{-4}

the per-mode foam average $\langle e^{i[\varphi_\nu(t+\tau)-\varphi_\nu(t)]}\rangle = e^{-D_\phi(\nu;\tau)/2}$; with $\langle|E_{\rm obs}|^2\rangle = \langle|E_{\rm src}|^2\rangle$ (unitary screen),

$$g^{(1)}_{\rm obs}(\tau) = \frac{\int d\nu\, S_{\rm src}(\nu)\, e^{-i2\pi\nu\tau}\, \mathcal{F}_{\rm foam}(\tau;\nu)}{\int d\nu\, S_{\rm src}(\nu)}, \qquad \mathcal{F}_{\rm foam}(\tau;\nu) = e^{-\sigma^2_\phi(\nu)[1-C_\alpha(\tau)]}. \tag{A4}$$

Spectrum and limiting cases

By Wiener–Khinchin (Eq. (22)), in the dynamical limit $C_\alpha \to 0$,

$$S_{\rm obs}(\nu) = e^{-\sigma^2_\phi(\nu)}\, S_{\rm src}(\nu) + (\text{pedestal}), \tag{A5}$$

the broadband core-plus-pedestal spectrum (38), for any $S_{\rm src}$. A blackbody gives the CMB spectrum (Section V B 2). For the Gaussian source $g^{(1)}_{\rm src}(\tau) = e^{-i\omega_0\tau}e^{-\tau^2/2\tau_G^2}$ (Eq. (19)), Eq. (A4) reproduces the closed form (21): carrier suppression $e^{-D_\phi/2}$, envelope broadening $\tau_G \to \sqrt{\tau_G^2+\sigma_\Delta^2}$ with $\sigma_\Delta^2 = (2\sigma_\ell^2/c^2)[1-C_\alpha]$ (Eq. (12)), at the single carrier ω_0.

Monte-Carlo validation

The closed form (21) is validated by simulating the shell model directly. Each realization draws $\Delta \sim \mathcal{N}(0,\sigma_\Delta^2)$ and forms the baseband coherence $B(\tau) = \langle e^{i\omega_0\Delta}\, e^{-(\tau-\Delta)^2/2\tau_G^2}\rangle$ over 4×10^6 realizations. With $\omega_0 = c = \tau_G = 1$, the simulation reproduces $\langle e^{i\omega_0\Delta}\rangle = e^{-\omega_0^2\sigma_\Delta^2/2}$ to machine precision and matches the closed form to Monte-Carlo noise; Table II reports the maximum deviation of $B(\tau)$ over $\tau \in [-3,3]\tau_G$ from the closed form (CF) and from the naive factorized form $\mathcal{F}_{\rm foam}\langle\gamma(\tau-\Delta)\rangle$ (FAC).